US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250282406
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	KAMINOKADO; Shingo et al.

CART

Abstract

A cart may include a body unit, a ground contact unit supported by the body unit, a prime mover configured to drive the ground contact unit, a first UWB anchor including a first antenna and a second antenna and configured to receive a beacon signal from a UWB tag, a second UWB anchor configured to receive the beacon signal from the UWB tag, and a control device configured to execute a following operation of causing the cart to follow the UWB tag by driving the prime mover. The control device may be configured to calculate a first distance which is a distance between the first UWB anchor and the UWB tag and a first tag angle which is an angle of the UWB tag relative to the first UWB anchor by using the beacon signal received by the first antenna and the beacon signal received by the second antenna.

Inventors: **KAMINOKADO; Shingo (Anjo-shi, JP), UMEMOTO; Ryo (Anjo-shi, JP), YAMAMOTO; Hirokatsu (Anjo-shi, JP), MATSUNO; Tadasuke (Anjo-shi, JP), OMONO; Hiroshi (Anjo-shi, JP)**

Applicant: **MAKITA CORPORATION** (Anjo-shi, JP)

Family ID: **96810168**

Assignee: **MAKITA CORPORATION (Anjo-shi, JP)**

Appl. No.: **19/060974**

Filed: **February 24, 2025**

Foreign Application Priority Data

JP	2024-037412	Mar. 11, 2024
JP	2024-037491	Mar. 11, 2024
JP	2024-037492	Mar. 11, 2024

Publication Classification

Int. Cl.: **B62B5/00** (20060101); **B62B5/06** (20060101)

U.S. Cl.:

CPC **B62B5/0069** (20130101); **B62B5/0063** (20130101); **B62B5/06** (20130101);

Background/Summary

REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority from Japanese Patent Application No. 2024-37491 filed on Mar. 11, 2024, Japanese Patent Application No. 2024-37492 filed on Mar. 11, 2024, and Japanese Patent Application No. 2024-37412 filed on Mar. 11, 2024. The entire content of the priority application is incorporated herein by reference.

TECHNICAL FIELD

[0002] The art disclosed herein relates to a cart.

BACKGROUND ART

[0003] Japanese Patent Application Publication No. 2023-13129 describes a cart. The cart includes: a body unit; a ground contact unit supported by the body unit and configured to be in contact with a ground; a prime mover configured to drive the ground contact unit; an anchor configured to receive a beacon signal from a tag configured to be carried by a user; and a control device configured to execute a following operation of causing the cart to follow the tag by driving the prime mover.

SUMMARY

[0004] In the cart configured to execute the following operation, precise calculation of a tag position, that is, a position of the user is desired. The present teachings provide an art configured to allow precise calculation of a position of a user.

[0005] A cart disclosed herein may comprise: a body unit; a ground contact unit supported by the body unit and configured to be in contact with a ground; a prime mover configured to drive the ground contact unit; a first UWB anchor including a first antenna and a second antenna different from the first antenna and configured to receive a beacon signal from a UWB tag configured to be carried by a user; a second UWB anchor a position of which is different from that of the first UWB anchor in a front-rear direction and configured to receive the beacon signal from the UWB tag; and a control device configured to execute a following operation of causing the cart to follow the UWB tag by driving the prime mover. The control device may be configured to calculate a first distance which is a distance between the first UWB anchor and the UWB tag and a first tag angle which is an angle of the UWB tag relative to the first UWB anchor by using the beacon signal received by the first antenna and the beacon signal received by the second antenna.

[0006] According to the above configuration, by using the first UWB anchor and the second UWB anchor with different positions in the front-rear direction, the position of the UWB tag, that is, the position of the user can be calculated with high precision.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. **1** illustrates a perspective view of a cart **2** according to a first embodiment from a right upper front side.

[0008] FIG. **2** illustrates a left view of the cart **2** according to the first embodiment.

[0009] FIG. **3** illustrates a perspective view of a body unit **4** according to the first embodiment from the right upper front side.

[0010] FIG. **4** illustrates a cross-sectional view of a right front overload detection mechanism **26**A according to the first embodiment from the right upper front side.
[0011] FIG. **5** illustrates a control configuration of the cart **2** according to the first embodiment.
[0012] FIG. **6** illustrates a right cross-sectional view of a housing **72** according to the first embodiment.
[0013] FIG. **7** illustrates a front view of a front UWB anchor **78** according to the first embodiment.
[0014] FIG. **8** illustrates a front cross-sectional view of a central switch box **98** according to the first embodiment.
[0015] FIG. **9** illustrates a right cross-sectional view of the central switch box **98** according to the first embodiment.
[0016] FIG. **10** illustrates a front view of a rear UWB anchor **108** according to the first embodiment.
[0017] FIG. **11** illustrates a flowchart of a following mode process according to the first embodiment.
[0018] FIG. **12** illustrates a situation where a UWB tag **200** is located in a following operation permitted area.
[0019] FIG. **13** illustrates a transmission path of a beacon signal when the UWB tag **200** is located in a following operation prohibited area.
[0020] FIG. **14** illustrates a situation where the UWB tag **200** is located in the following operation prohibited area.
[0021] FIG. **15** illustrates a schematic view of a cart **302** according to a reference example 1.
[0022] FIG. **16** illustrates a flowchart of a following mode process according to the reference example 1.
[0023] FIG. **17** illustrates a schematic diagram indicating a relation between the UWB tag **200** and a radio field intensity in the reference example 1.
[0024] FIG. **18** illustrates a schematic diagram of a cart **402** according to a reference example 2.
[0025] FIG. **19** illustrates a flowchart of a following mode process according to the reference example 2.
[0026] FIG. **20** illustrates a situation where the UWB tag **200** is located to the right of the cart **402** in the reference example 2.
[0027] FIG. **21** illustrates a situation where the UWB tag **200** is located to the left of the cart **402** in the reference example 2
[0028] FIG. **22** illustrates an overall perspective view of a cart **502** according to a second embodiment.
[0029] FIG. **23** illustrates a block diagram depicting a configuration of the cart **502** according to the second embodiment.
[0030] FIG. **24** illustrates an overall perspective view of a beacon **582** according to the second embodiment.
[0031] FIG. **25** illustrates a circuit diagram depicting an electrical configuration of the beacon **582** according to the second embodiment.
[0032] FIG. **26** illustrates a flowchart of processes executed by a microcontroller **602** of the beacon **582** when a main power of the beacon **582** according to the second embodiment is ON.
[0033] FIG. **27** illustrates a flowchart of processes executed by a control device **552** of the cart **502** when the main power of the cart **502** according to the second embodiment is ON and also a following mode is selected.
[0034] FIG. **28** illustrates a flowchart of a following operation control process executed by the control device **552** of the cart **502** according to the second embodiment.
[0035] FIG. **29** illustrates a diagram schematically depicting a first angular range A1, a second angular range A2, and a third angular range A3 as seen from the cart **502** according to the second embodiment.

[0036] FIG. **30** illustrates a graph indicating a relation between an offset angle θo and a turning curvature K during the following operation of the cart **502** in the cart **502** according to the second embodiment.
[0037] FIG. **31** is a flowchart of a following operation control process executed by a control device **552** of a cart **502** according to a third embodiment.

DESCRIPTION

[0038] Representative, non-limiting examples of the present disclosure will now be described in further detail with reference to the attached drawings. This detailed description is merely intended to teach a person of skill in the art further details for practicing aspects of the present teachings and is not intended to limit the scope of the present disclosure. Furthermore, each of the additional features and teachings disclosed below may be utilized separately or in conjunction with other features and teachings to provide improved carts as well as methods for using and manufacturing the same.
[0039] Moreover, combinations of features and steps disclosed in the following detailed description may not be necessary to practice the present disclosure in the broadest sense, and are instead taught merely to particularly describe representative examples of the present disclosure. Furthermore, various features of the above-described and below-described representative examples, as well as the various independent and dependent claims, may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings.
[0040] All features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter, independent of the compositions of the features in the embodiments and/or the claims. In addition, all value ranges or indications of groups of entities are intended to disclose every possible intermediate value or intermediate entity for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter.
[0041] A cart disclosed herein may comprise: a body unit; a ground contact unit supported by the body unit and configured to be in contact with a ground; a prime mover configured to drive the ground contact unit; a first UWB anchor including a first antenna and a second antenna different from the first antenna and configured to receive a beacon signal from a UWB tag configured to be carried by a user; a second UWB anchor a position of which is different from that of the first UWB anchor in a front-rear direction and configured to receive the beacon signal from the UWB tag; and a control device configured to execute a following operation of causing the cart to follow the UWB tag by driving the prime mover. The control device may be configured to calculate a first distance which is a distance between the first UWB anchor and the UWB tag and a first tag angle which is an angle of the UWB tag relative to the first UWB anchor by using the beacon signal received by the first antenna and the beacon signal received by the second antenna.
[0042] In one or more embodiments, the control device may be configured to calculate a second distance which is a distance between the second UWB anchor and the UWB tag by using the beacon signal received by the second UWB anchor. The control device may be configured to assess whether the UWB tag is located in a first area or a second area different from the first area by using the first distance and the second distance.
[0043] According to the above configuration, by using the first UWB anchor and the second UWB anchor with different positions in the front-rear direction, whether the user is located in the first area or in the second area can be determined with high precision.
[0044] In one or more embodiments, the first UWB anchor may be located frontward of the second UWB anchor. The first antenna and the second antenna may be aligned in a left-right direction. A midpoint between the first antenna and the second antenna in the left-right direction may match a center of the second UWB anchor in the left-right direction. When a second tag angle which is an

angle between a first virtual line connecting the first UWB anchor and the UWB tag and a second virtual line connecting the first UWB anchor and the second UWB anchor is an obtuse angle, the control device may determine that the UWB tag is located in the first area. When the second tag angle is 90 degrees or an acute angle, the control device may determine that the UWB tag is located in the second area.
[0045] There is a case where an area in which the user is located frontward of the first UWB anchor is the first area and an area in which the user is located directly laterally next to the first UWB anchor or rearward of the first UWB anchor is the second area. When the second tag angle is an obtuse angle, the user is located frontward of the first UWB anchor, and when the second tag angle is 90°, the user is directly laterally next to the first UWB anchor, and when the second tag angle is an acute angle, the user is located rearward of the first UWB anchor. Accordingly, whether the user is located in the first area or in the second area can be determined with high precision.
[0046] In one or more embodiments, the first area may be a following operation permitted area in which the following operation is permitted. The second area may be a following operation prohibited area in which the following operation is prohibited.
[0047] According to the above configuration, only when the user is located in the following operation permitted area, the cart can be caused to execute the following operation.
[0048] In one or more embodiments, when the control device determines that the UWB tag is located in the following operation permitted area, the control device may calculate a target velocity and a target angular velocity by using the first distance and the first tag angle to control the prime mover.
[0049] According to the above configuration, when the user is located in the following operation permitted area, the cart can be caused to operate so as to follow the user.
[0050] In one or more embodiments, when the control device determines that the UWB tag has moved from the following operation permitted area to the following operation prohibited area, the control device may stop the prime mover. When the control device determines that the UWB tag has moved from the following operation prohibited area to the following operation permitted area, the control device may drive the prime mover.
[0051] According to the above configuration, under a situation where the user is located in the following operation prohibited area, the cart can be suppressed from traveling. Accordingly, the cart can be caused to properly execute the following operation.
[0052] In one or more embodiments, the first UWB anchor may have a higher receiving sensitivity for signals arriving from a front side than for signals arriving from a rear side.
[0053] According to the above configuration, it becomes less likely that the beacon signal sent from a UWB tag located in the following operation prohibited area is received. In this case, when the beacon signal is not received, the control device can determine that the UWB tag is located in the following operation prohibited area. Accordingly, whether the UWB tag is located in the following operation prohibited area or not can be easily determined.
[0054] In one or more embodiments, the cart may further comprise a load platform supported by the body unit. One of the first UWB anchor and the second UWB anchor may be a front UWB anchor located frontward of the load platform.
[0055] A luggage/cargo, for example, is carried on the load platform. The beacon signal from the UWB tag may be interrupted by the luggage/cargo carried on the load platform, as a result of which the front UWB anchor cannot receive the beacon signal. According to the above configuration, because the beacon signal from the UWB tag is not interrupted by the luggage/cargo carried on the load platform, a possibility of the front UWB anchor receiving the beacon signal can be increased.
[0056] In one or more embodiments, the other of the first UWB anchor and the second UWB anchor may be a rear UWB anchor located rearward of the load platform. The rear UWB anchor may be located above the front UWB anchor and the load platform.
[0057] The beacon signal may not be received by the rear UWB anchor due to being interrupted by

the luggage/cargo carried on the load platform. According to the above configuration, because the beacon signal from the UWB tag is not interrupted by the luggage/cargo carried on the load platform, a possibility of the rear UWB anchor receiving the beacon signal from the UWB tag can be increased.

[0058] In one or more embodiments, the cart may further comprise a handle including a grip configured to be grasped by the user. The handle may be located above the load platform. The rear UWB anchor may be disposed in or on the handle.

[0059] According to the above configuration, the rear UWB anchor can be easily disposed above the load platform.

[0060] Another cart disclosed herein may be configured to execute a following operation of autonomously following a following target. The cart may comprise: a body; a wheel supported by the body and configured to be in contact with a ground; a prime mover configured to drive the wheel; an offset angle detector configured to detect an offset angle of the following target relative to a forward direction of the cart; and a control device. The control device is configured to execute a following operation control process of controlling the following operation of the cart. In the following operation control process, when the offset angle is within a first angular range including 0 degrees, the control device adjusts a turning degree of the cart during the following operation of the cart to a normal turning degree corresponding to the offset angle, and when the offset angle is within a second angular range adjacent to the first angular range, the control device adjusts the turning degree during the following operation of the cart to a reduced turning degree which is reduced from the normal turning degree corresponding to the offset angle.

[0061] When an absolute value of the offset angle is greater, a turning angle required to direct the forward direction of the cart toward the following target is greater. Due to this, increasing the turning degree of the cart may be considered. However, if the turning degree of the cart is excessively large, an inner wheel difference of the cart may be excessively enlarged. As a result, the cart may collide with an obstacle, as a result of which the following operation of the cart may not smoothly proceed. According to the above configuration, when the offset angle is within the second angular range (that is, the absolute value of the offset angle is relatively large), the turning degree of the cart is reduced than in normal times. Due to this, the turning degree of the cart can be suppressed from becoming excessively large, by which the inner wheel difference of the cart can be suppressed from being excessively enlarged. Due to this, the cart can be suppressed from contacting an obstacle, by which the following operation of the cart can be smoothly proceeded.

[0062] The “turning degree” herein is an index indicating slowness (or rapidness) of a turning motion the cart performs. The “turning degree” can be replaced with for example “a curvature (turning curvature) of a turning radius of the cart” or “a swiveling angle of a steering wheel relative to the forward direction of the cart”.

[0063] In one or more embodiments, the turning degree may include a turning curvature of the cart. In the following operation control process, the offset angle is within the first angular range, the control device may adjust the turning curvature during the following operation of the cart to a normal turning curvature corresponding to the offset angle, and when the offset angle is within the second angular range, the control device may adjust the turning curvature during the following operation of the cart to a reduced turning curvature which is reduced from the normal turning curvature corresponding to the offset angle.

[0064] According to the above configuration, when the offset angle is within the second angular range (i.e., the absolute value of the offset angle is relatively large), the turning curvature of the cart can be reduced than in normal times. Due to this, the turning curvature of the cart can be suppressed from being excessively increased, by which the inner wheel difference of the cart can be suppressed from being excessively enlarged. Due to this, since the cart can be suppressed from colliding with an obstacle, the following operation of the cart can be smoothly proceeded.

[0065] In one or more embodiments, the cart may be configured to switch between a following mode in which execution of the following operation is permitted and a manual mode in which the execution of the following operation is prohibited and the cart moves based on an operation from a user. A minimum turning radius of the cart in the following mode may be greater than a minimum turning radius of the cart in the manual mode.

[0066] When the cart has a smaller turning radius, the inner wheel difference of the cart is enlarged. Due to this, when the turning radius of the cart becomes small in the following mode in which the operation by the user does not intervene, the cart may contact an obstacle. In the meantime, in the manual mode in which the operation by the user intervenes, even when the turning radius of the cart becomes small to a certain degree, the possibility of the cart contacting an obstacle is probably low. Rather, if the turning radius of the cart cannot be made small in the manual mode, maneuverability of the cart may be decreased. According to the above configuration, the minimum turning radius of the cart in the following mode is greater than the minimum turning radius of the cart in the manual mode. As a result, the turning radius of the cart can be suppressed from being made small in the following mode, and the turning radius can be permitted to be small in the manual mode. Due to this, the cart can be suppressed from contacting an obstacle without imparting the maneuverability of the cart in the manual mode.

[0067] In one or more embodiments, in the following operation control process, when the offset angle is within a third angular range excluding the first angular range and the second angular range, the control device may stop the prime mover to stop the following operation of the cart.

[0068] When an absolute value of the offset angle is greater and thus a turning angle required to direct the forward direction of the cart to the following target is greater, the cart would need to be turned greatly. However, if the cart is turned greatly, the cart may contact an obstacle, by which the following operation of the cart may not proceed smoothly. According to the above configuration, when the offset angle is within the third angular range (i.e., the absolute value of the offset angle is relatively large), the following operation by the cart is stopped. Due to this, the cart can be suppressed from turning greatly. As a result, the cart can be suppressed from contacting the obstacle, the following operation of the cart can be smoothly proceeded.

[0069] Yet another cart disclosed herein may be configured to execute a following operation of autonomously following a following target. The cart may comprise: a body; a wheel supported by the body and configured to be in contact with a ground; a prime mover configured to drive the wheel; an offset angle detector configured to detect an offset angle of the following target relative to a forward direction of the cart; and a control device. The control device may be configured to operate the prime mover to cause the cart to execute the following operation when the offset angle is within an operation angular range including 0 degrees, and when the offset angle is within a stop angular range adjacent to the operation angular range, the control device may be configured to stop the prime mover to stop the following operation by the cart.

[0070] When an absolute value of the offset angle is greater and thus a turning angle required to direct the forward direction of the cart toward the following target is greater, the cart would need to be turned greatly. However, if the cart is turned greatly, the cart may contact an obstacle, by which the following operation of the cart may not proceed smoothly. According to the above configuration, when the offset angle is within the operation angular range (i.e., the absolute value of the offset angle is relatively small), the following operation by the cart is executed. When the offset angle is within the stop angular range (i.e., the absolute value of the offset angle is relatively large), the following operation by the cart is stopped. Due to this, the cart can be suppressed from turning greatly. As a result, the cart can be suppressed from contacting the obstacle, the following operation of the cart can be smoothly proceeded.

[0071] The present teachings further disclose a transport system. The transport system may comprise: a cart; and a communication terminal configured to communicate with the cart. The cart may be configured to execute a following operation of following the communication terminal when the communication terminal is moving. The communication terminal may comprise: a first operation part; and a second operation part located at a position different from the first operation part. When the first operation part is operated, the communication terminal may switch power of the communication terminal between ON and OFF. When the second operation part is operated during a predetermined start-operation receiving period with the power ON, the communication terminal may send a following start instruction which instructs to start the following operation to the cart.

[0072] According to the above configuration, the communication terminal includes the second operation part for causing the cart to start the following operation at the position different from the first operation part. Even if the first operation part is operated without user's intention, unless the second operation part is operated thereafter, the following operation by the cart is not started. Due to this, the following operation by the cart can be suppressed from being started without user's intension.

[0073] In one or more embodiments, the communication terminal may further comprise a third operation part disposed at a position different from those of the first operation part and the second operation part. When the third operation part is operated during a predetermined stop operation receiving period with the power ON, the communication terminal may send a following stop instruction which instructs to stop the following operation to the cart.

[0074] It is also conceivable to provide the communication terminal with a switching operation part configured to receive switching between starting and stopping of the following operation. In this configuration however, when the user operates the switching operation part, whether the communication terminal instructs the cart to start or stop the following operation may become unclear for the user. According to the above configuration, the operation part (second operation part) for receiving the start of the following operation and the operation part (third operation part) for receiving the stop of the following operation are disposed separate from each other. Due to this, when the user operates the second operation part/third operation part, whether the communication terminal instructs the cart to start/stop the following operation is clarified for the user.

[0075] In one or more embodiments, when the third operation part is operated during the stop operation receiving period with the power ON, the communication terminal may continuously send the following stop instruction to the cart until a predetermined send finish condition is satisfied.

[0076] For example, if communication connection is not good between the cart and the communication terminal, the following stop instruction sent from the communication terminal may not be received at the cart. If the communication terminal stops sending the following stop instruction despite the cart has not received the following stop instruction, the following operation by the cart would not be stopped, and thus would undesirably continue thereafter. In this configuration, a situation in which the user has operated the third operation part but the following operation by the cart is not stopped at all may happen. According to the above configuration, after the third operation part is operated, the communication terminal continuously sends the following stop instruction until the send finish condition is satisfied. Due to this, the situation in which the user has operated the third operation part but the following operation by the cart is not stopped at all can be suppressed.

[0077] In one or more embodiments, the send finish condition may include a first send finish condition that the second operation part is operated.

[0078] According to the above configuration, the communication terminal continuously sends the following stop instruction from when the third operation part is operated until the second operation part is operated next. Due to this, the situation in which the user has operated the third operation part but the following operation by the cart is not stopped at all can be suppressed.

[0079] In one or more embodiments, the cart may be configured to switch between a following mode in which execution of the following operation is permitted and a manual mode in which the execution of the following operation is prohibited and the cart moves based on an operation from the user. The start-operation receiving period may be at least a period during which the cart is in the

following mode.
[0080] If a period during which the cart is in the manual mode is the start-operation receiving period, when the second operation part is operated during the manual mode, the following operation by the cart may undesirably start at a timing when the cart is switched to the following mode. Due to this, the following operation by the cart may undesirably start a while late after the user has operated the second operation part. This means that the following operation by the cart starts at an unexpected timing for the user. According to the above configuration, the start-operation receiving period is at least a period during which the cart is in the following mode. Due to this, even if the second operation part is operated during the manual mode, since that operation is not accepted, the following operation by the cart is not to be started thereafter. Due to this, the following operation by the cart can be suppressed from being started at an unexpected timing for the user.
[0081] In one or more embodiments, the start-operation receiving period may be at least a period during which communication between the cart and the communication terminal is established.
[0082] If a period during which the communication between the cart and the communication terminal is lost is the start-operation receiving period, when the second operation part is operated during such period, the following operation by the cart may undesirably start at a timing when the communication is established thereafter. Due to this, the following operation by the cart may undesirably start a while late after the user has operated the second operation part. This means that the following operation by the cart starts at an unexpected timing for the user. According to the above configuration, the start-operation receiving period is at least a period during which the communication between the cart and the communication terminal is established. Due to this, even if the second operation part is operated during the period when the communication between the cart and the communication terminal is lost, since that operation is not accepted, the following operation by the cart is not to be started thereafter. Due to this, the following operation by the cart can be suppressed from being started at an unexpected timing for the user.
[0083] In one or more embodiments, the communication terminal may send a following stop instruction which instructs the cart to stop the following operation to the cart immediately after the power is turned ON.
[0084] According to the above configuration, the following operation by the cart is stopped immediately after the power of the communication terminal is turned ON. Due to this, for the first time after the second operation part is operated after the power of the communication terminal is turned ON, the following operation by the cart is started. Due to this, the following operation by the cart can be suppressed from being started without a user's intention.
[0085] The present teachings further disclose an electric apparatus. The electric apparatus may comprise: a microcontroller; a battery interface configured to be electrically connected to a battery; a first switch circuit configured to permit power supply to the microcontroller when the first switch circuit is ON and configured to prohibit power supply to the microcontroller when the first switch circuit is OFF; and a second switch circuit configured to permit power discharge from the battery when the second switch circuit is ON and prohibit power discharge from the battery when the second switch circuit is OFF.
[0086] When the electric apparatus is left unused for a long period of time with the battery attached to the battery interface, a remaining level of the battery may decrease due to natural discharge of electricity (natural power discharge). According to the above configuration, the second switch circuit configured to switch between permission/prohibition on the power discharge from the battery is provided separately from the first switch circuit configured to switch permission/prohibition on the power supply to the microcontroller (i.e., ON/OFF of power). In this configuration, by having the second switch circuit OFF, the natural power discharge from the battery can be suppressed. Due to this, by having the second switch circuit OFF when the electrical apparatus is left unused for a long period of time, the decrease in the remaining level of the battery

can be suppressed.
[0087] In one or more embodiments, the second switch circuit and the first switch circuit may be arranged in series on a power supply path from the battery interface to the microcontroller.
[0088] According to the above configuration, by having the second switch circuit OFF, irrespective of the ON/OFF state of the first switch circuit, the power supply to the microcontroller can be prohibited.
[0089] In one or more embodiments, the first switch circuit may be an electrical switch. The second switch circuit may be an electrical switch.
[0090] According to the above configuration, as compared to when the first switch circuit (second switch circuit) is for example a mechanical switch, the first switch circuit (second switch circuit) can be made smaller.
[0091] In one or more embodiments, the electrical apparatus may further comprise a USB port configured to receive a USB cable. The second switch circuit may be switched from OFF to ON when the electrical apparatus is supplied with power from the USB cable via the USB port.
[0092] According to the above configuration, with a relatively simple operation of connecting the USB cable to the USB port, the second switch circuit can be switched from OFF to ON.
[0093] In one or more embodiments, the battery may be a rechargeable secondary battery.
[0094] When the battery is a secondary battery, the battery may be sometimes configured difficult to be removed from the electrical apparatus. Due to this, the battery is stored in the state attached to the electrical apparatus, and natural power discharge from the battery may occur while the electrical apparatus is stored. In order to address this situation, according to the above configuration, by having the second switch circuit OFF while the electrical apparatus is stored, the natural power discharge from the battery can be suppressed.
[0095] (First Embodiment) A cart **2** illustrated in FIG. **1** comprises a body unit **4**, a load platform **6**, a handle unit **8**, and wheels **10**. The wheels **10** comprise a right front wheel **10**A, a left front wheel **10**B, a right rear wheel **10**C, and a left rear wheel **10**D (see FIG. **2**). The load platform **6**, the handle unit **8**, the right front wheel **10**A, the left front wheel **10**B, the right rear wheel **10**C, and the left rear wheel **10**D are all supported by the body unit **4**.
[0096] As illustrated in FIG. **3**, the body unit **4** comprises a first front frame **20**A, a first rear frame **20**B, a right frame **20**C, a left frame **20**D, a second front frame **22**A, and a second rear frame **22**B. The first front frame **20**A, the first rear frame **20**B, the second front frame **22**A, and the second rear frame **22**B all extend in a left-right direction. The second front frame **22**A is fixed to the first front frame **20**A from below at opposing ends of the second front frame **22**A in the left-right direction. The second rear frame **22**B is fixed to the first rear frame **20**B from below at opposing ends of the second rear frame **22**B in the left-right direction. The right frame **20**C and the left frame **20**D extend in a front-rear direction. The right frame **20**C and the left frame **20**D connect the first front frame **20**A and the first rear frame **20**B. The right frame **20**C is arranged to the right of a center of the first front frame **20**A in the left-right direction. The left frame **20**D is arranged to the left of the center of the first front frame **20**A in the left-right direction.
[0097] The body unit **4** further comprises a front auxiliary frame **24**A, a rear auxiliary frame **24**B, a right auxiliary frame **24**C, a left auxiliary frame **24**D, a handle support plate **24**E, and an overload detection mechanism **26**. The overload detection mechanism **26** comprises a right front overload detection mechanism **26**A, a left front overload detection mechanism **26**B, a right rear overload detection mechanism **26**C, and a left rear overload detection mechanism **26**D.
[0098] The front auxiliary frame **24**A, the rear auxiliary frame **24**B, the right auxiliary frame **24**C, and the left auxiliary frame **24**D are disposed below the second front frame **22**A and the second rear frame **22**B. The front auxiliary frame **24**A and the rear auxiliary frame **24**B extend in the left-right direction. The right auxiliary frame **24**C and the left auxiliary frame **24**D extend in the front-rear direction. The right auxiliary frame **24**C connects a right end of the front auxiliary frame **24**A and a right end of the rear auxiliary frame **24**B. A front portion of the right auxiliary frame **24**C is

attached to the second front frame **22**A via the right front overload detection mechanism **26**A. A rear portion of the right auxiliary frame **24**C is attached to the second rear frame **22**B via the right rear overload detection mechanism **26**C. The left auxiliary frame **24**D connects a left end of the front auxiliary frame **24**A and a left end of the rear auxiliary frame **24**B. A front portion of the left auxiliary frame **24**D is attached to the second front frame **22**A via the left front overload detection mechanism **26**B. A rear portion of the left auxiliary frame **24**D is attached to the second rear frame **22**B via the left rear overload detection mechanism **26**D.

[0099] The right front overload detection mechanism **26**A, the left front overload detection mechanism **26**B, the right rear overload detection mechanism **26**C and the left rear overload detection mechanism **26**D all have a same configuration. Hereafter, the right front overload detection mechanism **26**A only will be described, and descriptions for the left front overload detection mechanism **26**B, the right rear overload detection mechanism **26**C, and the left rear overload detection mechanism **26**D will be omitted.

[0100] As illustrated in FIG. **4**, the right front overload detection mechanism **26**A comprises a base part **30**, a buffer member **31**, a housing part **32**, a cap part **34**, a shaft **36**, a coil spring **38**, a detection plate **40**, and a detection sensor **42**.

[0101] The base part **30** comprises a first cylindrical part **30**A, a first flange part **30**B extending outward from an outer peripheral surface of the first cylindrical part **30**A, and an inner projection part **30**C extending inward from an inner peripheral surface of a lower end of the first cylindrical part **30**A. Existence of the inner projection part **30**C defines a first hole **30**D having a smaller diameter than an inner diameter of the first cylindrical part **30**A. The buffer member **31** is attached to an upper portion of the base part **30**.

[0102] The housing part **32** comprises a second cylindrical part **32**A and a second flange part **32**B extending outward from an outer peripheral surface of an upper end of the second cylindrical part **32**A. An inner diameter of an upper portion of the second cylindrical part **32**A is greater than an outer diameter of the first cylindrical part **30**A of the base part **30**. A first frame hole **28**A is defined in the front portion of the right auxiliary frame **24**C. An outer diameter of the second cylindrical part **32**A is smaller than a diameter of the first frame hole **28**A. The second cylindrical part **32**A penetrates the first frame hole **28**A in an up-down direction. A diameter of the second flange part **32**B is the same as a diameter of the first flange part **30**B of the base part **30**. The diameter of the second flange part **32**B is greater than a diameter of the first frame hole **28**A. A lower surface of the first flange part **30**B abuts an upper surface of the second flange part **32**B. An upper surface of the right auxiliary frame **24**C abuts a lower surface of the second flange part **32**B. The cap part **34** is screwed from below to the second cylindrical part **32**A.

[0103] The shaft **36** comprises a first shaft part **36**A, a second shaft part **36**B extending downward from a lower end of the first shaft part **36**A, and a third shaft part **36**C extending downward from a lower end of the second shaft part **36**B. A second frame hole **28**B is defined in a right end of the second front frame **22**A. The first shaft part **36**A penetrates the second frame hole **28**B in the up-down direction. A diameter of the second shaft part **36**B is greater than a diameter of the first shaft part **36**A. The second shaft part **36**B is configured to slide inside the first cylindrical part **30**A of the base part **30** in the up-down direction. A diameter of the third shaft part **36**C is smaller than the diameter of the second shaft part **36**B. An upper bolt hole **36**D is defined in an upper portion of the shaft **36**, and a lower bolt hole **36**E is defined in a lower portion of the shaft **36**. A first bolt **44** to which a spacer **46** is attached is screwed into the upper bolt hole **36**D. A second bolt **48** is screwed into the lower bolt hole **36**E via the detection plate **40**. The coil spring **38** is disposed between the third shaft part **36**C and the first cylindrical part **30**A of the base part **30**. An upper end of the coil spring **38** abuts a lower surface of the second shaft part **36**B, and a lower end of the coil spring **38** abuts an upper surface of the inner projection part **30**C of the base part **30**.

[0104] The detection plate **40** comprises a disk part **40**A having a second hole **40**B at its center and a detector part **40**C. The detector part **40**C extends downward from opposing ends of the disk part

40A in the front-rear direction. The disk part **40**A is disposed between the second bolt **48** and the third shaft part **36**C of the shaft **36** in the up-down direction. The second bolt **48** secures the detection plate **40** to the shaft **36**. That is, the shaft **36** and the detection plate **40** move as a unit.
[0105] The detection sensor **42** is a so-called photo interrupter. The detection sensor **42** comprises a light emitting element **50** and a light receiving element **52** disposed to face each other. The detection sensor **42** is off when there is no interruption between the light emitting element **50** and the light receiving element **52**, and is on when there is an interruption between the light emitting element **50** and the light receiving element **52**. The detection sensor **42** is electrically connected to a main control device **60** (see FIG. **5**). The main control device **60** of FIG. **5** applies a pulsing voltage to the light emitting element **50**. Specifically, the main control device **60** switches between a state of applying a voltage to the light emitting element **50** and a state of not applying the voltage to the light emitting element **50** at predetermined cycles. Due to this, the light emitting element **50** intermittently emits light, i.e., blinks.
[0106] As illustrated in FIG. **4**, in a state where there is no load on the load platform **6** and thus no load from the load platform **6** is applied on the shaft **36**, due to biasing force of the coil spring **38**, an upper surface of the disk part **40**A of the detection plate **40** abuts a lower surface of the inner projection part **30**C of the base part **30**. In this state, the detector part **40**C of the detection plate **40** does not intervene between the light emitting element **50** and the light receiving element **52**. Due to this, the light receiving element **52** receives the intermittent light from the light emitting element **50**. In this case, the detection sensor **42** sends an on signal and an off signal at the predetermined cycles. Thus, when the main control device **60** of FIG. **5** receives the on signal and the off signal at the predetermined cycles from the detection sensor **42**, the main control device **60** determines that there is no overload.
[0107] When, starting from the state illustrated in FIG. **4**, a luggage/cargo is carried on the load platform **6** and thus a load from the load platform **6** is applied on the second front frame **22**A, the shaft **36** and the detection plate **40** move downward relative to the base part **30** against the biasing force of the coil spring **38**. At this occasion, when a load of a predetermined upper limit load or more is applied on the shaft **36**, the detector part **40**C of the detection plate **40** is positioned between the light emitting element **50** and the light receiving element **52**. Due to this, the detector part **40**C intervenes the light from the light emitting element **50**. Thus, the light receiving element **52** does not receive light. In this case, the detection sensor **42** continuously sends the off signal to the main control device **60** (see FIG. **5**). Also, when a power wiring connected to the light emitting element **50** is disconnected, the light emitting element **50** does not emit light, and the light receiving element **52** does not receive the light. In this case also, the detection sensor **42** continuously sends the off signal to the main control device **60** (see FIG. **5**). Due to this, when the main control device **60** of FIG. **5** receives the off signal from the detection sensor **42**, the main control device **60** determines that an overload is occurring or the power wiring is disconnected.
[0108] Further, if a connector connected to the light emitting element **50** of FIG. **4** is experiencing a short circuit, the light emitting element **50** emits continuous light instead of the intermittent light. In this case, the light receiving element **52** receives the continuous light from the light emitting element **50** and the detection sensor **42** continuously sends the on signal to the main control device **60** (see FIG. **5**). When the main control device **60** of FIG. **5** continuously receives the on signal from the detection sensor **42**, the main control device **60** determines that the connector for example is experiencing a short circuit.
[0109] As an assumption, a case where the light emitting element **50** is emitting continuous light in a normal state where disconnection and/or short circuit are not taking place will be described. In this case, the light emitting element **50** emits the continuous light when there is no overload and when a short circuit is taking place. Due to this, when the main control device **60** receives the on signal continuously from the detection sensor **42**, the main control device **60** cannot differentiate which situation is occurring, i.e., there is no overload or a short circuit is taking place. According to

the above configuration, the main control device **60** is able to accurately distinguish when there is no overload, when there is overload, or when some sort of abnormality is taking place.
[0110] As illustrated in FIG. **1**, the cart **2** further comprises a bumper **70**, a housing **72**, a right front light **73**A, and a left front light **73**B all attached to a front portion of the body unit **4**. As illustrated in FIG. **2**, the bumper **70** and the housing **72** are disposed in front of the load platform **6**. The housing **72** is disposed above the bumper **70**. A front end of the housing **72** is disposed rearward of a front end of the bumper **70**. The right front light **73**A is disposed on the right side of the bumper **70**. The left front light **73**B is disposed on the left side of the bumper **70**.
[0111] As illustrated in FIG. **6**, the housing **72** is composed of a first front housing **74** and a first rear housing **76** fixed to the first front housing **74**. The housing **72** accommodates a front UWB anchor **78** therein. The front UWB anchor **78** conforms to a UWB (Ultra Wide Band) standard. In the left-right direction, a center of the front UWB anchor **78** in the left-right direction matches a position of the cart **2**. As illustrated in FIG. **7**, the front UWB anchor **78** comprises a UWB substrate **80**, a first antenna **82**, a second antenna **84**, an anchor control device **86**, and a shield **88** (see FIG. **6**). The first antenna **82**, the second antenna **84**, and the anchor control device **86** are disposed on a front surface of the UWB substrate **80**. The first antenna **82** and the second antenna **84** are disposed above the anchor control device **86**. The first antenna **82** and the second antenna **84** are aligned in the left-right direction. A midpoint between the first antenna **82** and the second antenna **84** in the left-right direction matches a center of the cart **2** and a center of the front UWB anchor **78**. The first antenna **82** is disposed to the right of the center of the UWB substrate **80** in the left-right direction. The second antenna **84** is disposed to the left of the center of the UWB substrate **80** in the left-right direction. As illustrated in FIG. **6**, the shield **88** is disposed on a rear surface of the UWB substrate **80**. The shield **88** is a member configured to shield a signal transported from a rear side of the front UWB anchor **78**. The shield **88** makes a receiving sensitivity for signals arriving from a front side of the front UWB anchor **78** smaller than for signals arriving from the rear side of the front UWB anchor **78**.
[0112] As illustrated in FIG. **1**, the handle unit **8** comprises a fixing shaft **90**, a coupling member **92**, a support shaft **94**, a handle part **96**, a central switch box **98**, and a right switch box **100**. The fixing shaft **90** is fixed to the handle support plate **24**E of the body unit **4**, and extends upward from the handle support plate **24**E. The support shaft **94** extends in the up-down direction. The support shaft **94** is disposed rearward of the fixing shaft **90**. The support shaft **94** is pivotably coupled to the fixing shaft **90** via the coupling member **92**.
[0113] As illustrated in FIG. **8**, the handle part **96** is attached to an upper end of the support shaft **94**. As illustrated in FIG. **1**, the handle part **96** comprises a right grip **96**A and a left grip **96**B. The central switch box **98** is arranged at a center portion of the handle part **96** in the left-right direction. As illustrated in FIG. **9**, the central switch box **98** is composed of a second front housing **102**, a second rear housing **104** fixed to the second front housing **102**, and an upper plate **106** sandwiched between the second front housing **102** and the second rear housing **104**. The central switch box **98** accommodates the rear UWB anchor **108**. The rear UWB anchor **108** conforms to the UWB standard. That is, the center of the rear UWB anchor **108** in the left-right direction matches the center of the front UWB anchor **78** (see FIG. **3**) in the left-right direction. As illustrated in FIG. **10**, the rear UWB anchor **108** comprises a UWB substrate **110**, a third antenna **112**, a fourth antenna **114**, an anchor control device **116**, and a shield **118** (see FIG. **9**). The third antenna **112**, the fourth antenna **114**, and the anchor control device **116** are disposed on a front surface of the UWB substrate **110**. The third antenna **112** and the fourth antenna **114** are disposed below the anchor control device **116**. The third antenna **112** and the fourth antenna **114** are aligned in the left-right direction. A midpoint between the third antenna **112** and the fourth antenna **114** in the left-right direction matches the center of the cart **2** and a center of the rear UWB anchor **108**. The third antenna **112** is disposed to the right of a center of the UWB substrate **110** in the left-right direction. The fourth antenna **114** is disposed to the left of the center of the UWB substrate **110** in the left-

right direction. As illustrated in FIG. **9**, the shield **118** is disposed on a rear surface of the UWB substrate **110**. The shield **118** is a member configured to shield a signal transported from the rear side of the rear UWB anchor **108**. The shield **118** makes a receiving sensitivity for signals arriving from the front side of the rear UWB anchor **108** smaller than a receiving sensitivity for signals arriving from the rear side of the front UWB anchor **78**. As illustrated in FIG. **2**, the central switch box **98** is disposed above the housing **72** and the load platform **6**. That is, the rear UWB anchor **108** inside the central switch box **98** is disposed above the front UWB anchor **78** in the housing **72** and the load platform **6**.

[0114] As illustrated in FIG. **1**, the right switch box **100** is disposed between the right grip **96**A and the central switch box **98**. A tail light **101** is disposed at a lower portion of the right switch box **100**. As illustrated in FIG. **5**, the right switch box **100** further comprises a main power switch **120**, a moving direction switch **122**, and a speed switch **124**. The main power switch **120** is configured to switch on/off of a main power of the cart **2**. The moving direction switch **122** is configured to switch a moving direction of the cart **2** in the manual mode. The speed switch **124** is configured to switch a moving speed of the cart **2** in the manual mode. The central switch box **98** includes a mode switch **126** and an emergency stop switch **128**. The mode switch **126** is configured to switch an operation mode of the cart **2** between the manual mode, a following mode, and a parking mode. The emergency stop switch **128** is configured to stop traveling of the cart **2**.

[0115] The user can operate to pivot the handle unit **8** while the user is grasping the right grip **96**A and the left grip **96**B of the handle unit **8** in FIG. **1** with his/her both hands. As illustrated in FIG. **5**, the cart **2** comprises a handle angle sensor **130** configured to detect a pivoting angle of the handle unit **8** as a handle angle, a steering mechanism **132** configured to steer the right front wheel **10**A and the left front wheel **10**B as steering wheels, and a steering motor **134** configured to drive the steering mechanism **132**. The steering motor **134** is for example a brushless motor. The handle angle sensor **130**, the steering mechanism **132**, and the steering motor **134** are supported by the body unit **4** (see FIG. **1**).

[0116] As illustrated in FIG. **1**, the cart **2** comprises a battery receptacle part **136** arranged on the body unit **4**. The battery receptacle part **136** is configured to have a battery pack **138** (see FIG. **5**) detachably attached thereto. The battery pack **138** comprises for example secondary battery cell(s) (not illustrated) such as lithium-ion battery cell(s), and is rechargeable by using a charger (not illustrated). The cart **2** operates on power supplied from the battery pack **138** attached to the battery receptacle part **136**.

[0117] As illustrated in FIG. **5**, the cart **2** comprises a traction motor **140**. The traction motor **140** comprises a right front wheel motor **140**A configured to drive the right front wheel **10**A, a left front wheel motor **140**B configured to drive the left front wheel **10**B, a right rear wheel motor **140**C configured to drive the right rear wheel **10**C, and a left rear wheel motor **140**D configured to drive the left rear wheel **10**D. The right front wheel motor **140**A, the left front wheel motor **140**B, the right rear wheel motor **140**C, and the left rear wheel motor **140**D are for example brushless motors. The right front wheel motor **140**A, the left front wheel motor **140**B, the right rear wheel motor **140**C, and the left rear wheel motor **140**D are supported by the body unit **4** (see FIG. **1**).

[0118] The cart **2** further comprises a control power circuit **150**. The control power circuit **150** is configured to permit supply of power from the battery pack **138** when an on operation is performed on the main power switch **120**, and prohibit the supply of power from the battery pack **138** when an off operation is performed on the main power switch **120**. The main control device **60** controls operation of the cart **2**. The main control device **60** controls operations of the right front wheel motor **140**A, the left front wheel motor **140**B, the right rear wheel motor **140**C, the left rear wheel motor **140**D, and the steering motor **134** via motor drivers **160**, **162**, **164**, **166**, and **168**. Although this is not illustrated, a brake circuit is connected to the motor drivers **160**, **162**, **164**, and **166** corresponding to the right front wheel motor **140**A, the left front wheel motor **140**B, the right rear wheel motor **140**C, and the left rear wheel motor **140**D. The main control device **60** can exert great braking force on the right front wheel motor **140**A, the left front wheel motor **140**B, the right rear wheel motor **140**C, and the left rear wheel motor **140**D by applying great current to the break circuit while the right front wheel motor **140**A, the left front wheel motor **140**B, the right rear wheel motor **140**C, and the left rear wheel motor **140**D are rotating. The control power circuit **150**, the main control device **60**, the motor drivers **160**, **162**, **164**, **166**, and **168**, and the break circuit are supported by the body unit **4** (see FIG. **1**).

[0119] Further, the four detection sensors **42**, the front UWB anchor **78**, and the rear UWB anchor **108** are electrically connected to the main control device **60**.

[0120] The cart **2** of FIG. **1** is configured to operate in the manual mode, the following mode, or the parking mode. In the manual mode, the cart **2** moves frontward or rearward according to a user's operation with the user standing behind the body unit **4** grasping the handle unit **8** with his/her hands. In the following mode, the cart **2** moves following the UWB tag **200** (see FIG. **12**) the user standing in front of the body unit **4** is carrying. In the parking mode, the cart **2** continues to remain stopped without receiving a command from the handle unit **8** or a command from the UWB tag **200**. Here, the main control device **60** of the cart **2** is configured to prohibit the cart **2** from traveling when the main control device **60** determines that an overload is occurring or some sort of abnormality is occurring in the cart **2** based on the information received from the detection sensor(s) **42**. That is, the traveling of the cart **2** in the manual mode or the following mode is prohibited. Further, the cart **2** is put on emergency stop when the emergency stop switch **128** (see FIG. **5**) is operated.

[0121] (Following Mode Process; FIG. **11**) With reference to FIG. **11**, a following mode process executed by the main control device **60** will be described. The main control device **60** starts processes of FIG. **11** when the operation mode of the cart **2** is set to the following mode.

[0122] As illustrated in FIG. **12**, when the front UWB anchor **78** receives a beacon signal sent from the UWB tag **200** via the first antenna **82** and the second antenna **84** (see FIG. **7**), the front UWB anchor **78** calculates a distance from the front UWB anchor **78** to the UWB tag **200** (hereafter, “first distance D1”) and an angle of the UWB tag **200** relative to the front UWB anchor **78** (hereafter, “first angle a1”). The first angle a1 is an angle between a line L0 extending in the front-rear direction and a first line L1 connecting the front UWB anchor **78** and the UWB tag **200**. Specifically, the front UWB anchor **78** specifies the first distance D1 by using a transmission time of the beacon signal received via the first antenna **82** and calculates the first angle a1 by using a phase difference between the beacon signal received via the first antenna **82** and the beacon signal received via the second antenna **84**. As a method of calculating an angle by using a phase difference, Phase Differences of Arrival (PDoA), Angle of Arrival (AoA) are known, for example.

[0123] When the rear UWB anchor **108** receives the beacon signal sent from the UWB tag **200** via the third antenna **112** and the fourth antenna **114** (see FIG. **10**), the rear UWB anchor **108** calculates a distance from the rear UWB anchor **108** to the UWB tag **200** (hereafter, “second distance D2”) and an angle of the UWB tag **200** relative to the rear UWB anchor **108** (hereafter, “second angle a2”). The second angle a2 is an angle between the line L0 extending in the front-rear direction and a second line L2 connecting the rear UWB anchor **108** and the UWB tag **200**. Calculation methods of the second distance D2 and the second angle a2 are respectively the same as the calculation methods of the first distance D1 and the first angle a1.

[0124] In S**10** of FIG. **11**, the main control device **60** obtains the first distance D1 and the first angle a1 from the front UWB anchor **78**. Specifically, the main control device **60** obtains the first distance D1 and the first angle a1 from the front UWB anchor **78** in response to supplying a first obtaining instruction for obtaining the first distance D1 and the first angle a1 to the front UWB anchor **78**. Here, when the front UWB anchor **78** has not received the beacon signal from the UWB tag **200**, the main control device **60** receives not-receiving information indicating that the beacon signal has not been received from the front UWB anchor **78**.

[0125] In S**12**, the main control device **60** obtains the second distance D2 and the second angle a2 from the rear UWB anchor **108**. Specifically, the main control device **60** obtains the second distance D2 and the second angle a2 from the rear UWB anchor **108** in response to supplying a second obtaining instruction for obtaining the second distance D2 and the second angle a2 to the rear UWB anchor **108**. When the rear UWB anchor **108** has not received the beacon signal from the UWB tag **200**, the main control device **60** obtains the not-receiving information indicating that the beacon signal has not been received from the rear UWB anchor **108**.

[0126] In S**20**, the main control device **60** determines whether the UWB tag **200** is located in a following operation permitted area. The following operation permitted area is an area in which execution of a following operation of moving by following the UWB tag **200** is permitted. As illustrated in FIG. **12**, the following operation permitted area is an area frontward of the front UWB anchor **78**. Further, an area directly laterally next to the front UWB anchor **78** and rearward of the front UWB anchor **78** is a following operation prohibited area in which the execution of the following operation is prohibited. The main control device **60** determines whether the UWB tag **200** is located in the following operation permitted area or not by using the first distance D1 obtained from the front UWB anchor **78**, the second distance D2 obtained from the rear UWB anchor **108**, and a third distance D3 between the front UWB anchor **78** and the rear UWB anchor **108**. Here, the third distance D3 is pre-stored in a memory (not illustrated) of the main control device **60**. The main control device **60** calculates a third angle a3 by using the first distance D1, the second distance D2, the third distance D3, and a following Formula (1). The third angle a3 is an angle between the first line L1 and a third line L3 connecting the front UWB anchor **78** and the rear UWB anchor **108**. In the present embodiment, the third line L3 and the line L0 are parallel to each other.

[Formula1]

[00001]
$$\cos(a3) = \frac{D1^2 + D3^2 - D2^2}{2 \times D1 \times D3} \quad (1)$$

[0127] With reference to FIG. **12**, a case where the UWB tag **200** is located frontward of the front UWB anchor **78** will be described. In this case, in response to the second distance D2 being longer than the first distance D1, the third angle a3 is an obtuse angle. For this reason, when the third angle a3 is an obtuse angle, it can be determined that the UWB tag **200** is located in the following operation permitted area.

[0128] With reference to FIGS. **13** and **14**, a case where the UWB tag **200** is located rearward of the front UWB anchor **78** will be described. In the present embodiment, the front UWB anchor **78** and the rear UWB anchor **108** respectively have the shield **88** (see FIG. **6**) and a shield **118** (see FIG. **9**). Due to this, as illustrated in FIG. **13**, the first antenna **82** and the second antenna **84** of the front UWB anchor **78** and the third antenna **112** and the fourth antenna **114** of the rear UWB anchor **108** do not receive much of the beacon signals sent from the rear sides of the front UWB anchor **78** and the rear UWB anchor **108** (broken-line arrows in FIG. **13**). However, the beacon signal(s) from the rear sides of the front UWB anchor **78** and the rear UWB anchor **108** may turn around to the front sides of the front UWB anchor **78** and the rear UWB anchor **108** and thus reach the first antenna **82** and the second antenna **84** of the front UWB anchor **78** and the third antenna **112** and the fourth antenna **114** of the rear UWB anchor **108** (bold-line arrows in FIG. **13**). FIG. **14** illustrates a situation where the beacon signals sent from the rear sides of the front UWB anchor **78** and the rear UWB anchor **108** turn around to the front sides of the front UWB anchor **78** and the rear UWB anchor **108** and reach the first antenna **82** and the second antenna **84** of the front UWB anchor **78** and the third antenna **112** and the fourth antenna **114** of the rear UWB anchor **108**, in a more understandable form. In this case, as illustrated in FIG. **14**, in response to the first distance D1 being longer than the second distance D2, the third angle a3 is an acute angle. Due to this, by using the third angle a3, it can be specified that the UWB tag **200** is located in the following operation prohibited area. Also when the UWB tag **200** is located frontward of the front UWB anchor **78** and also rearward of the rear UWB anchor **108**, the third angle a3 is an acute angle.

[0129] Back to FIG. **11**, in S**20**, when the third angle a3 is an obtuse angle, the main control device **60** determines that the UWB tag **200** is located frontward of the front UWB anchor **78**. In this case, the main control device **60** determines that the UWB tag **200** is located in the following operation permitted area (YES to S**20**), and the process proceeds to S**22**. Contrary to this, when the third angle a3 is not an obtuse angle, the main control device **60** determines that the UWB tag **200** is not located frontward of the front UWB anchor **78**. In this case, the main control device **60** determines that the UWB tag **200** is located in the following operation prohibited area (NO to S**20**), and the process proceeds to S**30**.
[0130] When in S**10** the not-receiving information has been received from the front UWB anchor **78** or when in S**12** the not-receiving information has been received from the rear UWB anchor **108**, the main control device **60** determines that the UWB tag **200** is located in the following operation prohibited area (NO to S**20**), and the process proceeds to S**30**. When the not-receiving information is obtained, it is likely that the UWB tag **200** is located rearward of the front UWB anchor **78** or the rear UWB anchor **108**.
[0131] In S**22**, the main control device **60** determines a target velocity TV and a target angular velocity Tω of the cart **2** so that the cart **2** follows the UWB tag **200** by using the first distance D1 and the first angle a1 that are obtained from the front UWB anchor **78**.
[0132] In S**24**, the main control device **60** controls the operation of the cart **2**. Specifically, the main control device **60** controls the operations of the right front wheel motor **140**A, the left front wheel motor **140**B, the right rear wheel motor **140**C, the left rear wheel motor **140**D, and the steering motor **134** based on the target velocity TV and the target angular velocity Tω. When S**24** ends, the process returns to S**10**. For example, when the UWB tag **200** moves from the following operation prohibited area to the following operation permitted area while the cart **2** remains stopped, the main control device **60** controls the operations of the right front wheel motor **140**A, the left front wheel motor **140**B, the right rear wheel motor **140**C, the left rear wheel motor **140**D, and the steering motor **134** so that the cart **2** travels.
[0133] In S**30**, the main control device **60** decides the target velocity TV and the target angular velocity Tω of the cart **2** to be zero. When S**30** ends, the process proceeds to S**24**.
[0134] In S**24** after S**30**, the main control device **60** controls the operation of the cart **2** so that the cart **2** does not travel. For example, when the UWB tag **200** moves from the following operation permitted area to the following operation prohibited area while the cart **2** is traveling, the main control device **60** controls operation of the brake circuit so that the cart **2** stops. Also when the UWB tag **200** moves from the following operation permitted area to the following operation prohibited area while the cart **2** remains stopped, the main control device **60** does not drive the right front wheel motor **140**A, the left front wheel motor **140**B, the right rear wheel motor **140**C, the left rear wheel motor **140**D, and the steering motor **134**.
[0135] As mentioned above, the main control device **60** can accurately determine whether the UWB tag **200** is located in the following operation permitted area or not by using the first distance D1, the second distance D2, and the third distance D3. Accordingly, the following operation can be suppressed from being executed when the UWB tag **200** is located in the following operation prohibited area.
[0136] Hereafter, the main control device **60** and the anchor control devices **86**, **116** will be collectively referred to as “control unit”.
[0137] In one or more embodiments, the cart **2** comprises: the body unit **4**; the wheels **10** (example for “ground contact part”) supported by the body unit **4** and configured to be in contact with a ground; the traction motor **140** (example for “prime mover”) configured to drive the wheels **10**; the front UWB anchor **78** (example for “first UWB anchor”) including the first antenna **82** and the second antenna **84** and configured to receive the beacon signal from the UWB tag **200** configured to be carried by a user; the rear UWB anchor **108** (example for “second UWB anchor”) disposed rearward of the front UWB anchor **78** and configured to receive the beacon signal from the UWB

tag **200**; and the control unit (example for “control device”) configured to execute the following operation of causing the cart **2** to follow the UWB tag **200** by driving the traction motor **140**. The control unit is configured to calculate the first distance D1 between the front UWB anchor **78** and the UWB tag **200** and the first angle a1 (example for “a first tag angle”) of the UWB tag **200** relative to the front UWB anchor **78** by using the beacon signal received by the first antenna **82** and the beacon signal received by the second antenna **84**.

[0138] According to the above configuration, by using the front UWB anchor **78** and the rear UWB anchor **108** with different positions in the front-rear direction, the position of the UWB tag **200**, that is, the position of the user can be calculated with high precision.

[0139] In one or more embodiments, the control unit is configured to calculate the second distance D2 between the rear UWB anchor **108** and the UWB tag **200** by using the beacon signal received by the rear UWB anchor **108**, and configured to determine whether the UWB tag **200** is located in the following operation permitted area (example for “a first area”) in which execution of the following operation is permitted or the following operation prohibited area (example for “a second area”) in which the execution of the following operation is prohibited by using the first distance D1 and the second distance D2.

[0140] According to the above configuration, by using the front UWB anchor **78** and the rear UWB anchor **108** with different positions in the front-rear direction, whether the user is located in the following operation permitted area or in the following operation prohibited area can be determined with high precision. Also, according to the above configuration, only when the user is located in the following operation permitted area, the cart **2** can be caused to execute the following operation.

[0141] In one or more embodiments, the front UWB anchor **78** is located frontward of the rear UWB anchor **108**. The first antenna **82** and the second antenna **84** are aligned in the left-right direction. The midpoint between the first antenna **82** and the second antenna **84** in the left-right direction matches the center of the rear UWB anchor **108**. When the third angle a3 (example for “second tag angle”) between the first line L1 (example for “a first virtual line”) connecting the front UWB anchor **78** and the UWB tag **200** and the third line L3 (example for “a second virtual line”) connecting the front UWB anchor **78** and the rear UWB anchor **108** is an obtuse angle, the control unit determines that the UWB tag **200** is located in the following operation permitted area, and when the third angle a3 is 90 degrees or an acute angle, the control unit determines that the UWB tag **200** is located in the following operation prohibited area.

[0142] There is a case where an area in which the user is located frontward of the front UWB anchor **78** is the following operation permitted area and an area in which the user is located directly laterally next to the front UWB anchor **78** or rearward of the front UWB anchor **78** is the following operation prohibited area. When the third angle a3 is an obtuse angle, the user is located frontward of the front UWB anchor **78**, when the third angle a3 is 90°, the user is directly laterally next to the front UWB anchor **78**, and when the third angle a3 is an acute angle, the user is located rearward of the front UWB anchor **78**. Accordingly, whether the user is located in the following operation permitted area or in the following operation prohibited area can be determined with high precision.

[0143] In one or more embodiments, when the control unit determines that the UWB tag **200** is located in the following operation permitted area, the control unit calculates the target velocity TV and the target angular velocity Tω by using the first distance D1 and the first angle a1 to control the traction motor **140**.

[0144] According to the above configuration, when the user is located in the following operation permitted area, the cart **2** can be caused to operate so as to follow the user.

[0145] In one or more embodiments, when the control unit determines that the UWB tag **200** has moved from the following operation permitted area to the following operation prohibited area, the control unit stops the traction motor **140**, and when the control unit determines that the UWB tag **200** has moved from the following operation prohibited area to the following operation permitted area, the control unit drives the traction motor **140**.

[0146] According to the above configuration, under a situation where the user is located in the following operation prohibited area, the cart **2** can be suppressed from traveling. Accordingly, the cart **2** can be caused to properly execute the following operation.
[0147] In one or more embodiments, the front UWB anchor **78** has a higher receiving sensitivity for signals arriving from the front side than for signals arriving from the rear side.
[0148] According to the above configuration, it becomes less likely that the beacon signal sent from the UWB tag **200** located in the following operation prohibited area is received. In this case, when the beacon signal is not received, the control unit can determine that the UWB tag **200** is located in the following operation prohibited area. Accordingly, whether the UWB tag **200** is located in the following operation prohibited area or not can be easily determined.
[0149] In one or more embodiments, the cart **2** further comprises the load platform **6** supported by the body unit **4**. The front UWB anchor **78** is located frontward of the load platform **6**.
[0150] A luggage/cargo is carried on the load platform **6**. When the UWB tag is located on the front side of the cart **2** and the front UWB anchor **78** is located rearward of the load platform **6**, the beacon signal from the UWB tag **200** may be interrupted by the luggage/cargo carried on the load platform **6**, as a result of which the front UWB anchor **78** cannot receive the beacon signal. According to the above configuration, because the beacon signal from the UWB tag **200** is not interrupted by the luggage/cargo carried on the load platform **6**, a possibility of the front UWB anchor **78** receiving the beacon signal can be increased.
[0151] In one or more embodiments, the rear UWB anchor **108** is located rearward of the load platform **6**. The rear UWB anchor **108** is located above the front UWB anchor **78** and the load platform **6**.
[0152] The beacon signal from the UWB tag **200** may not be received by the rear UWB anchor **108** due to being interrupted by the load platform **6**. According to the above configuration, because the beacon signal from the UWB tag **200** is not interrupted by the luggage/cargo carried on the load platform **6**, a possibility of the rear UWB anchor **108** receiving the beacon signal from the UWB tag **200** can be increased.
[0153] In one or more embodiments, the cart **2** further comprises the handle part **96** including the grips **96**A, **96**B configured to be grasped by the user. The handle part **96** is located above the load platform **6** and the rear UWB anchor **108** is disposed in or on the handle part **96**.
[0154] According to the above configuration, the rear UWB anchor **108** can be easily disposed above the load platform **6**.
[0155] (First Modification) One of the front UWB anchor **78** and the rear UWB anchor **108** may include one antenna only. In the present modification, the main control device **60** controls operation of the cart **2** by using a distance and an angle that are calculated by the one of the front UWB anchor **78** and the rear UWB anchor **108** which includes the two antennas.
[0156] (Second Modification) In S**22** of FIG. **11**, the control unit may determine the target velocity TV and the target angular velocity Tω of the cart **2** by using the second distance D2 and the second angle a2 that are obtained from the rear UWB anchor **108**.
[0157] (Third Modification) In the left-right direction, the center of the front UWB anchor **78** and the center of the rear UWB anchor **108** may not match. In this case, the control unit executes the process of S**20** in FIG. **11** by correcting a discrepancy between the center of the front UWB anchor **78** and the center of the rear UWB anchor **108**.
[0158] (Fourth Modification) The control unit may determine whether the UWB tag **200** is located in the following operation permitted area or not by using the first distance D1 and the second distance D2. For example, when the first distance D1 is longer than the second distance D2, the control unit determines that the UWB tag **200** is located in the following operation permitted area.
[0159] (Fifth Modification) Also when the operation mode of the cart **2** is the manual mode or the parking mode, the control unit may specify whether the UWB tag **200** is located in the following operation permitted area or not.

[0160] (Sixth Modification) At least one of or both of the front UWB anchor **78** and the rear UWB anchor **108** may not comprise the shield **88**, **118**.
[0161] (Seventh Modification) Both of the front UWB anchor **78** and the rear UWB anchor **108** may be disposed frontward of the load platform **6** or may be disposed rearward of the load platform **6**. Further, at least one of the front UWB anchor **78** and the rear UWB anchor **108** may be disposed on a lateral side of the load platform **6**.
[0162] (Eighth Modification) The "prime mover" may not be limited to the traction motor **140**, and may be an engine, for example.
[0163] (Ninth Modification) The "ground contact part" may not limited to wheel(s), but may be crawler(s), for example.
[0164] (Tenth Modification) An area frontward of the front UWB anchor **78** may be a forward following area in which a following operation of causing the cart **2** to move forward and following the UWB tag **200** is executed and an area rearward of the front UWB anchor **78** may be a retract travel area in which a following operation of causing the cart **2** to retract rearward and following the UWB tag **200** is executed. In this case, an area directly laterally next to the front UWB anchor **78** is preferably a stop area in which the following operation is not executed. In the present modification, the "forward following area" and the "retract following area" are an example of "first area" and "second area", respectively.
[0165] (Eleventh Modification) An area frontward of the front UWB anchor **78** may be a front lighting area in which the right front light **73**A and the left front light **73**B both light and an area rearward of the front UWB anchor **78** may be a rear lighting area in which the tail light **101** lights. In this case, the area directly laterally next to the front UWB anchor **78** is preferably a light off area in which the right front light **73**A, the left front light **73**B, and the tail light **101** all do not light. In the present modification, the "front lighting area" and the "rear lighting area" are an example of "first area" and "second area", respectively.
[0166] Some of further features of the cart **2** disclosed in the present embodiment will be listed hereinbelow.
(Feature 1-1)
[0167] A detector comprising: [0168] a light emitting element; [0169] a light receiving element disposed to face the light emitting element; [0170] a shield plate configured to shield light emitted by the light emitting element; and [0171] a control unit, [0172] wherein the control unit applies a pulsing voltage on the light emitting element so that the light emitting element intermittently emits light, [0173] the light receiving element intermittently receives the light emitted by the light emitting element when the shield plate does not shield the light emitted by the light emitting element, and [0174] the light receiving element does not receive the light emitted by the light emitting element when the shield plate shields the light emitted by the light emitting element.
(Feature 1-2)
[0175] The detector according to feature 1-1, wherein the light receiving element continuously receives the light emitted by the light emitting element when a short circuit abnormality is occurring in the light emitting element and the shield plate does not shield the light emitted by the light emitting element.
(Feature 1-3)
[0176] A cart comprising: [0177] a load platform; [0178] a ground contact part configured to be in contact with a ground; [0179] a prime mover configured to drive the ground contact part; [0180] a detector attached to the load platform and according to feature 1-1 or 1-2; and [0181] a control unit configured to control operation of the prime mover, [0182] wherein the control unit permits to drive the prime mover when the light receiving element of the detector receives intermittently the light emitted by the light emitting element, and the control unit prohibits driving of the prime mover when the light receiving element does not receive the light emitted by the light emitting element.

(Feature 1-4)
[0183] The cart according to feature 1-3, wherein when a load of a predetermined upper limit load or more is applied from the load platform on the shied plate of the detector, the light emitted by the light emitting element of the detector is shielded by the shield plate, and [0184] when a load less than the predetermined upper limit load is applied from the load platform on the shield plate of the detector, the light emitted by the light emitting element of the detector is not shielded by the shield plate.
[0185] Effects brought by the above-mentioned features 1-1 to 1-4 will be described.
[0186] In one or more embodiments, the overload detection mechanism **26** (example for “detector”) comprises: the light emitting element **50**; the light receiving element **52** disposed to face the light emitting element **50**; the detection plate **40** (example for “shield plate”) configured to shield the light emitted by the light emitting element **50**, and the control unit. The control unit applies a pulsing voltage on the light emitting element **50** so that the light emitting element **50** intermittently emits light. The light receiving element **52** intermittently receives the light emitted by the light emitting element **50** when the detection plate **40** does not shield the light emitted by the light emitting element **50**, and the light receiving element **52** does not receive the light emitted by the light emitting element **50** when the detection plate **40** shields the light emitted by the light emitting element **50**.
[0187] According to the above configuration, the control unit can detect whether the detection plate **40** is shielding the light emitted by the light emitting element **50** according to whether the light receiving element **52** is intermittently receiving the light emitted by the light emitting element **50** or not.
[0188] In one or more embodiments, the light receiving element **52** continuously receives the light emitted by the light emitting element **50** when a short circuit abnormality is occurring in the light emitting element **50** and also the detection plate **40** does not shield the light emitted by the light emitting element **50**.
[0189] According to the above configuration, the control unit can detect that a short circuit abnormality is occurring in the light emitting element **50** when the light receiving element **52** continuously receives the light emitted by the light emitting element **50**.
[0190] In one or more embodiments, the cart **2** comprises: the load platform **6**; the wheels **10** (example for “ground contact part”) configured to be in contact with a ground; the traction motor **140** (example for “prime mover”) configured to drive the wheels **10**; the overload detection mechanism **26** (example for “detector”) attached to the load platform **6**; and the control unit configured to control operation of the traction motor **140**. The control unit permits to drive the traction motor **140** when the light receiving element **52** receives intermittently the light emitted by the light emitting element **50**, and the control unit prohibits the traction motor **140** when the light receiving element **52** does not receive the light emitted by the light emitting element **50**.
[0191] When the light receiving element **52** does not receive the light emitted by the light emitting element **50**, it is likely that some sort of abnormality is occurring in the cart **2**. According to the above configuration, when some sort of abnormality is occurring in the cart **2**, the cart **2** can be suitably suppressed from traveling.
[0192] In one or more embodiments, when a load of a predetermined upper limit load or more is applied from the load platform **6** on the detection plate **40** (example for “shield plate”), the light emitted by the light emitting element **50** is shielded by the detection plate **40**, and when a load less than the predetermined upper limit load is applied from the load platform **6** on the detection plate **40**, the light emitted by the light emitting element **50** is not shielded by the detection plate **40**.
[0193] A case where the light emitting element **50** is emitting continuous light under a normal state with no abnormality such as disconnection or short-circuit occurring will be described. In this case, when there is no overload and short circuit is occurring, the light emitting element **50** continuously emits light. Due to this, the main control device **60** cannot distinguish whether there is no overload

or there is a short-circuit when the main control device **60** continuously receives the ON signal from the detection sensor **42**. According to the above configuration, the overload detection mechanism **26** enables to determine precisely whether there is overload or not. Accordingly, when there is overload, the cart **2** can be suitably suppressed from traveling.
[0194] (Modification 1-1) The “detector” can be implemented in a robotic cleaner, not only in the cart **2**.
[0195] (Reference Example 1) With reference to FIGS. **15** to **17**, a cart **302** according to reference example 1 will be described. Hereafter, mainly, differences between the cart **302** of the reference example 1 and the cart **2** of the first embodiment will be described, and descriptions of like points will be omitted.
[0196] As illustrated in FIG. **15**, the cart **302** of the reference example 1 differs from the cart **2** of the first embodiment in that the cart **302** does not have the rear UWB anchor **108**. Further, the main control device **60** (see FIG. **5**) in the reference example 1 executes the following mode process of FIG. **16** instead of the following mode process of FIG. **12**.
[0197] (Following Mode Process; FIG. **16**) With reference to FIG. **16**, the following mode process executed by the main control device **60** will be described. The main control device **60** starts processes of FIG. **16** when the operation mode of the cart **302** is set to the following mode.
[0198] In S**110**, the main control device **60** obtains the first distance D1 and the first angle a1 from the front UWB anchor **78**. Specifically, in response to supplying a first obtaining instruction for obtaining the first distance D1 and the first angle a1 to the front UWB anchor **78**, the main control device **60** obtains the first distance D1 and the first angle a1 from the front UWB anchor **78**. Here, when the front UWB anchor **78** has not received the beacon signal from the UWB tag **200**, the main control device **60** obtains not-receiving information indicating that the beacon signal has not been received from the front UWB anchor **78**.
[0199] In S**112**, the main control device **60** determines a radio field intensity threshold by using the first distance D1 obtained in S**110**. The radio field intensity threshold is a threshold for determining whether the UWB tag **200** is located in the following operation permitted area or not. The main control device **60** sets a smaller radio field intensity threshold when the first distance D1 is longer.
[0200] In S**120** of FIG. **16**, the main control device **60** determines whether the UWB tag **200** is located in the following operation permitted area or not. Specifically, the main control device **60** determines whether the radio field intensity of the beacon signal received by the front UWB anchor **78** exceeds the radio field intensity threshold determined in S**112** or not. With reference to FIG. **17**, the radio field intensity of the beacon signal will be described. In the present reference example, the front UWB anchor **78** has the shield **88** (see FIG. **6**). Due to this, as illustrated in FIG. **17**, the first antenna **82** and the second antenna **84** of the front UWB anchor **78** does not receive much of the beacon signals sent from the rear side of the front UWB anchor **78** (broken-line arrow in FIG. **17**). However, a beacon signal from the rear side of the front UWB anchor **78** may turn around to the front side of the front UWB anchor **78** and reach the first antenna **82** and the second antenna **84** (solid-line arrow in FIG. **17**). Because the beacon signal turns around to the front side of the front UWB anchor **78**, the radio field intensity of this beacon signal is low. Due to this, the radio field intensity of the beacon signal which turns from the rear side of the front UWB anchor **78** over to the front side of the front UWB anchor **78** and reaches the first antenna **82** and the second antenna **84** has a lower radio field intensity than that of the beacon signal which reaches the first antenna **82** and the second antenna **84** from the front side of the front UWB anchor **78**.
[0201] Back to FIG. **16**, when the radio field intensity exceeds the radio field intensity threshold, the main control device **60** determines that the UWB tag **200** is located in the following operation permitted area (YES to S**120**), the process proceeds to S**122**. Contrary to this, when the radio field intensity is equal to or lower than the radio field intensity threshold, the main control device **60** determines that the UWB tag **200** is located in the following operation prohibited area (NO to S**120**), and the process proceeds to S**140**.

[0202] When in S**110** the main control device **60** has received the not-receiving information from the front UWB anchor **78**, the main control device **60** determines that the UWB tag **200** is located in the following operation prohibited area (NO to S**120**), and the process proceeds to S**140**. This is because, when the not-receiving information is obtained, it is likely that the UWB tag **200** is located rearward of the front UWB anchor **78**.
[0203] In S**122**, the main control device **60** calculates a tentative target velocity TTV and a target angular velocity Tω by using the first distance D1 and the first angle a1 so that the cart **2** follows the UWB tag **200**. Specifically, the main control device **60** calculates the tentative target velocity TTV by using the first distance D1 and a following Formula (2), and calculates the target angular velocity Tω by using the first angle a1 and a following Formula (3). "Vg1", "Vo1" in the Formula (2) are respectively a first velocity gain and a first velocity offset. The first velocity offset Vo1 is a value for stopping the cart **302** in front of the user. "ωg1" of the Formula (3) is a first angular velocity gain.

[00002] [Formula2] [Formula3]

$$\mathrm{TTV} = (D1 \times \mathrm{Vg1}) - \mathrm{Vo1} \quad (2) \qquad T = a1 \times \omega \mathrm{g1} \quad (3)$$

[0204] In S**124**, the main control device **60** calculates a limit velocity VL by using the first angle a1. Specifically, the main control device **60** calculates the limit velocity VL by using a following Formula (4). "Vmax", "Vo2" in the Formula (4) are a maximum velocity and a second velocity offset of the cart **302**. The second velocity offset Vo2 is a value determined according to the first angle a1. The second velocity offset Vo2 increases as the first angle a1 increases. That is, as the first angle a1 increases, the limit velocity VL decreases.

[00003] [Formula4]

$$\mathrm{VL} = V\mathrm{max} - \mathrm{Vo2} \quad (4)$$

[0205] In S**126**, the main control device **60** determines whether the tentative target velocity TTV exceeds the limit velocity VL or not. When the main control device **60** determines that the tentative target velocity TTV exceeds the limit velocity VL (YES to S**126**), the process proceeds to S**128**. Contrary to this, when the main control device **60** determines that the tentative target velocity TTV does not exceed the limit velocity VL (NO to S**126**), the process proceeds to S**130**.
[0206] In S**128**, the main control device **60** decides the limit velocity VL as the target velocity TV.
[0207] Further, in S**130**, the main control device **60** decides the tentative target velocity TTV as the target velocity TV.
[0208] In S**132**, the main control device **60** controls operation of the cart **302**. Specifically, the main control device **60** controls operations of the right front wheel motor **140**A, the left front wheel motor **140**B, the right rear wheel motor **140**C, the left rear wheel motor **140**D, and the steering motor **134** based on the target velocity TV and the target angular velocity Tω. When S**132** ends, the process returns to S**110**.
[0209] Also, in S**140**, the main control device **60** decides the target velocity TV and the target angular velocity Tω of the cart **302** to be zero. When S**140** ends, the process proceeds to S**132**.
[0210] In S**132** after S**140**, the main control device **60** prohibits the cart **302** from traveling.
[0211] As mentioned above, the main control device **60** can accurately determine whether the UWB tag **200** is located in the following operation permitted area or not by using the radio field intensity and the radio field intensity threshold. Accordingly, when the UWB tag **200** is located in the following operation prohibited area, the following operation can be suppressed from being executed.
[0212] Some of features of the cart **302** disclosed in the present reference example will be listed hereinbelow.
(Feature 2-1)
[0213] A cart comprising: [0214] a body unit; [0215] a ground contact part supported by the body unit and configured to be in contact with a ground; [0216] a prime mover configured to drive the

ground contact part; [0217] a UWB anchor configured to receive a beacon signal from a UWB tag configured to be carried by a user; and [0218] a control unit, [0219] wherein the UWB anchor has a higher receiving sensitivity for signals arriving from a front side than for signals arriving from a rear side, and [0220] the control unit determines whether the UWB tag is located frontward of the UWB anchor or not by using a radio field intensity of the beacon signal received from the UWB tag.
(Feature 2-2)
[0221] The cart according to feature 2-1, wherein the UWB anchor comprises a first antenna and a second antenna different from the first antenna, and [0222] the control unit calculates a distance between the UWB anchor and the UWB tag and an angle of the UWB tag relative to the UWB anchor by using the beacon signal received by the first antenna and the beacon signal received by the second antenna.
(Feature 2-3)
[0223] The cart according to feature 2-1 or feature 2-2, wherein the control unit determines that the UWB tag is located frontward of the UWB anchor when the radio field intensity exceeds the radio field intensity threshold, and [0224] the control unit determines that the UWB tag is located rearward of the UWB anchor when the radio field intensity is equal to or lower than the radio field intensity threshold.
(Feature 2-4)
[0225] The cart according to feature 2-3, wherein the control unit calculates the distance between the UWB tag and the UWB anchor by using the beacon signal received by the UWB anchor, and sets the radio field intensity threshold to be smaller as the distance is longer.
(Feature 2-5)
[0226] The cart according to any one of features 2-1 to 2-4, further comprising a load platform supported by the body unit, [0227] wherein the UWB anchor is disposed frontward of the load platform.
(Feature 2-6)
[0228] The cart according to any one of features 2-1 to 2-5, wherein the cart comprises only one UWB anchor.
(Feature 2-7)
[0229] The cart according to any one of features 2-1 to 2-6, [0230] wherein the control unit determines that the UWB tag is located in a following operation permitted area in which moving to follow the UWB tag is permitted when the control unit determines that the UWB tag is located frontward of the UWB anchor, [0231] the control unit determines that the UWB tag is located in the following operation prohibited area in which moving to follow the UWB tag is prohibited when the control unit determines that the UWB tag is located rearward of the UWB anchor, and [0232] the control unit stops operation of the prime mover when the control unit determines that the UWB tag has moved from the following operation permitted area to the following operation prohibited area.
[0233] Some of effects by the above-described features 2-1 to 2-7 will be described.
[0234] In one or more embodiments, the cart **302** comprises: the body unit **4**, the wheels **10** (example for “ground contact part”) supported by the body unit **4** and configured to be in contact with a ground; the traction motor **140** (example for “prime mover”) configured to drive the wheels **10**; the front UWB anchor **78** (example for “UWB anchor”) configured to receive a beacon signal from the UWB tag **200** configured to be carried by a user; and the control unit. The front UWB anchor **78** has a higher receiving sensitivity for signals arriving from the front side than for signals arriving from the rear side. The control unit determines whether the UWB tag **200** is located frontward of the front UWB anchor **78** or not by using the radio field intensity of the beacon signal received from the UWB tag **200**.
[0235] According to the above configuration, by utilizing the radio field intensity, whether the

UWB tag **200**, i.e., the user is located frontward of the front UWB anchor **78** or not can be acknowledged.
[0236] In one or more embodiments, the front UWB anchor **78** comprises the first antenna **82** and the second antenna **84**. The control unit calculates the first distance D1 between the front UWB anchor **78** and the UWB tag **200** and the first angle a1 of the UWB tag **200** relative to the front UWB anchor **78** by using the beacon signal received by the first antenna **82** and the beacon signal received by the second antenna **84**.
[0237] According to the above configuration, the position of the user can be more accurately grasped.
[0238] In one or more embodiments, the control unit determines that the UWB tag **200** is located frontward of the front UWB anchor **78** when the radio field intensity exceeds the radio field intensity threshold, and the control unit determines that the UWB tag **200** is located rearward of the front UWB anchor **78** when the radio field intensity is equal to or lower than the radio field intensity threshold.
[0239] According to the above configuration, whether the user is located frontward of the front UWB anchor **78** can be more accurately grasped.
[0240] In one or more embodiments, the control unit calculates the first distance D1 between the UWB tag **200** and the front UWB anchor **78** by using the beacon signal received by the front UWB anchor **78**, and sets the radio field intensity threshold to be smaller as the first distance D1 is longer.
[0241] The longer the distance between the UWB tag **200** and the front UWB anchor **78**, the smaller the radio field intensity. Accordingly, by making the radio field intensity threshold smaller as the first distance D1 is longer, irrespective of the distance between the UWB tag **200** and the front UWB anchor **78**, the user can grasp whether the user is located frontward of the front UWB anchor **78**.
[0242] In one or more embodiments, the cart **302** further comprises the load platform **6** supported by the body unit **4**. The front UWB anchor **78** is arranged frontward of the load platform **6**.
[0243] A luggage/cargo, for example, is carried on the load platform **6**. When the UWB tag is located on the front side of the cart **2** and also the front UWB anchor **78** is located rearward of the load platform **6**, the beacon signal from the UWB tag **200** may be interrupted by the load carried on the load platform **6**, as a result of which the front UWB anchor **78** cannot receive the beacon signal. According to the above configuration, since the beacon signal from the UWB tag **200** is not interrupted by the luggage/cargo carried on the load platform **6**, a possibility of the front UWB anchor **78** receiving the beacon signal can be increased. Also, because the receiving sensitivity of the front UWB anchor **78** for the front side is higher than the receiving sensitivity thereof for the rear side, by utilizing the radio field intensity, whether the user is located frontward of the front UWB anchor **78** or not can be accurately grasped.
[0244] n one or more embodiments, the cart **302** comprises only one front UWB anchor **78**.
[0245] According to the above configuration, as compared to the configuration where two or more UWB anchors are used to grasp whether the user is located frontward of the front UWB anchor **78** or not, the configuration of the cart **302** can be simplified.
[0246] In one or more embodiments, the control unit determines that the UWB tag **200** is located in the following operation permitted area in which moving to follow the UWB tag **200** is permitted when the control unit determines that the UWB tag **200** is located frontward of the front UWB anchor **78**, the control unit determines that the UWB tag **200** is located in the following operation prohibited area in which moving to follow the UWB tag **200** is prohibited when the control unit determines that the UWB tag **200** is located rearward of the front UWB anchor **78**, and the control unit stops operation of the traction motor **140** when the control unit determines that the UWB tag **200** has moved from the following operation permitted area to the following operation prohibited area.

[0247] According to the above configuration, the cart **302** can be suppressed from traveling when the user is located in the following operation prohibited area. Accordingly, the cart **302** can be caused to suitably execute the following operation.
[0248] (Modification 2-1) The cart **302** of the reference example 1 may comprise two or more UWB anchors. In this case, at least one of the two or more UWB anchors may comprise two antennas. Specifically, the UWB anchor for detecting the radio field intensity and the UWB anchor for detecting a distance to a tag may be different.
[0249] (Modification 2-2) The radio field intensity threshold may be constant irrespective of how long the distance between the UWB tag **200** and the front UWB anchor **78** is.
[0250] (Modification 2-3) The front UWB anchor **78** of the reference example 1 may be disposed rearward of the load platform **6**.
[0251] (Modification 2-4) The control unit may determine that the UWB tag **200** is located in a forward following area in which a following operation of causing the cart **302** to move forward and following the UWB tag **200** is executed when the control unit determines that the UWB tag **200** is located frontward of the front UWB anchor **78**, and the control unit may determine that the UWB tag **200** is located in a retreat following area in which a following operation of causing the cart **302** to retreat and follow the UWB tag **200** is executed when the control unit determines that the UWB tag **200** is located rearward of the front UWB anchor **78**.
[0252] (Modification 2-5) The control unit may determine that the UWB tag **200** is located in a front lighting area in which the right front light **73**A and the left front light **73**B light when the control unit determines that the UWB tag **200** is located frontward of the front UWB anchor **78**, and may determine that the UWB tag **200** is located in a rear lighting area in which the tail light **101** lights when the control unit determines that the UWB tag **200** is located rearward of the front UWB anchor **78**.
[0253] (Reference Example 2) With reference to FIGS. **18** to **21**, a cart **402** according to a reference example 2 will be described. Hereafter, mainly, differences between the cart **402** of the reference example 2 and the cart **2** of the first embodiment will be described, and descriptions for like points will be omitted.
[0254] As illustrated in FIG. **18**, the cart **402** of the reference example 2 differs from the cart **2** of the first embodiment in that the cart **402** of the reference example 2 comprises a right-front UWB anchor **412** and a left-front UWB anchor **410** instead of the front UWB anchor **78** (see FIG. **7**) and the rear UWB anchor **108** (see FIG. **10**). Also, the main control device **60** (see FIG. **5**) of the reference example 2 executes a following mode process of FIG. **19** instead of the following mode process of FIG. **12**.
[0255] The left-front UWB anchor **410** and the right-front UWB anchor **412** in FIG. **18** have a same configuration as the front UWB anchor **78** of the first embodiment (see FIGS. **6**, **7**). The left-front UWB anchor **410** and the right-front UWB anchor **412** are arranged in the housing **72** (see FIG. **1**) of the cart **402**. That is, the left-front UWB anchor **410** and the right-front UWB anchor **412** are arranged frontward of the load platform **6** (see FIG. **1**). The left-front UWB anchor **410** is arranged to the left of the center of the cart **402** in the left-right direction. The right-front UWB anchor **412** is located to the right of the center of the cart **402** in the left-right direction. In the left-right direction, a midpoint between the two antennas of the left-front UWB anchor **410** and a midpoint between the two antennas of the right-front UWB anchor **412** are arranged at equal intervals from the center of the cart **2**.
[0256] (Following Mode Process; FIG. **19**) With reference to FIG. **19**, the following mode process executed by the main control device **60** will be described. The main control device **60** starts processes of FIG. **19** when the operation mode of the cart **402** is set to the following mode.
[0257] As illustrated in FIG. **20**, when the left-front UWB anchor **410** receives the beacon signal from the UWB tag **200**, the left-front UWB anchor **410** calculates a distance from the left-front UWB anchor **410** to the UWB tag **200** (hereafter, "fourth distance D4"). Also, when the right-front

UWB anchor **412** receives the beacon signal from the UWB tag **200**, the right-front UWB anchor **412** calculates a distance from the right-front UWB anchor **412** to the UWB tag **200** (hereafter, “fifth distance D5”). A method of calculating the fourth distance D4, the fifth distance D5 is the same as the method of calculating the first distance D1 of the first embodiment.

[0258] In S**210** in FIG. **19**, the main control device **60** obtains the fourth distance D4 from the left-front UWB anchor **410**. Specifically, the main control device **60** obtains the fourth distance D4 from the left-front UWB anchor **410** in response to supplying a third obtaining instruction for obtaining the fourth distance D4 to the left-front UWB anchor **410**. Here, when the left-front UWB anchor **410** has not received the beacon signal from the UWB tag **200**, the main control device **60** obtains the not-receiving information indicating that the beacon signal has not been received from the left-front UWB anchor **410**.

[0259] In S**212**, the main control device **60** obtains the fifth distance D5 from the right-front UWB anchor **412**. Specifically, the main control device **60** obtains the fifth distance D5 from the right-front UWB anchor **412** in response to supplying a fourth obtaining instruction for obtaining the fifth distance D5 to the right-front UWB anchor **412**. Here, when the right-front UWB anchor **412** has not received the beacon signal from the UWB tag **200**, the main control device **60** obtains the not-receiving information indicating that the beacon signal has not been received from the right-front UWB anchor **412**.

[0260] In S**214**, the main control device **60** determines whether the UWB tag **200** is located in the following operation permitted area or not. Specifically, the main control device **60** determines whether the not-receiving information has been received from at least one of the left-front UWB anchor **410** and the right-front UWB anchor **412** or not. When the main control device **60** determines that the not-receiving information has not been received, the main control device **60** determines that the UWB tag **200** is located frontward of the left-front UWB anchor **410** and the right-front UWB anchor **412**. In this case, the main control device **60** determines that the UWB tag **200** is located in the following operation permitted area (YES to S**214**), and the process proceeds to S**220**. Contrary to this, when the main control device **60** determines that the not-receiving information has been received, the control device **60** determines that the UWB tag **200** is located rearward of the left-front UWB anchor **410** and the right-front UWB anchor **412**. In this case, the main control device **60** determines that the UWB tag **200** is located in the following operation prohibited area (YES to S**214**), and the process proceeds to S**216**.

[0261] In S**216**, the main control device **60** decides the target velocity TV and the target angular velocity Tω of the cart **402** to be zero, and stops the cart **302**. When S**216** ends, the process returns to S**210**.

[0262] In S**220**, the main control device **60** assesses whether the fourth distance D4 and the fifth distance D5 match. When the main control device **60** determines that the fourth distance D4 and the fifth distance D5 match (YES to S**220**), the process proceeds to S**222**. Here, when the fourth distance D4 and the fifth distance D5 match is when the UWB tag **200** is located right in front of the cart **402**. Contrary to this, when the main control device **60** determines that the fourth distance D4 and the fifth distance D5 do not match (NO to S**220**), the process proceeds to S**230**.

[0263] In S**222**, the main control device **60** calculates the target velocity TV of the cart **402**. Specifically, the main control device **60** calculates the target velocity TV by using the fourth distance D4, the fifth distance D5, and a following Formula (5). “DA” and “Vg2” in the Formula (5) are respectively an average value of the fourth distance D4 and the fifth distance D5 and a second velocity gain.

[00004] [Formula5]

$$TV = DA \times Vg2 \quad (5)$$

[0264] In S**224**, the main control device **60** causes the cart **402** to move straight by using the target velocity TV. When S**224** ends, the process returns to S**210**.

[0265] Also, in S**230**, the main control device **60** determines whether the fourth distance D4 is longer than the fifth distance D5. When the main control device **60** determines that the fourth distance D4 is longer than the fifth distance D5 (YES to S**230**), the process proceeds to S**232**. Here, when the fourth distance D4 is longer than the fifth distance D5 is when, as in FIG. **20**, the UWB tag **200** is located to the right of the center of the cart **402** in the left-right direction. Contrary to this, when the main control device **60** determines that the fourth distance D4 is not longer than the fifth distance D5 (NO to S**230**), the process proceeds to S**240**. Here, when the fourth distance D4 is not longer than the fifth distance D5 is when, as in FIG. **21**, the UWB tag **200** is located to the left of the center of the cart **402** in the left-right direction.
[0266] As illustrated in FIG. **19**, in S**232**, the main control device **60** calculates the target velocity TV and the target angular velocity Tω of the cart **402** so that the cart **402** follows the UWB tag **200**. A method of calculating the target velocity TV in the present step is the same as the method of calculating the target velocity TV in S**222**. The main control device **60** calculates the target angular velocity Tω by using the fourth distance D4, the fifth distance D5, and a following Formula (6). "ωg2" in the Formula (6) is a second angular velocity gain.

[Formula6]

[00005] $T = .Math. D4 - D5 .Math. \times g2 \quad (6)$

[0267] In S**234**, the main control device **60** causes the cart **402** to turn right by using the target velocity TV and the target angular velocity Tω. When S**234** ends, the process returns to S**210**.
[0268] In S**240**, the main control device **60** calculates the target velocity TV and the target angular velocity Tω of the cart **402** so that the cart **402** follows the UWB tag **200**. A method of calculating the target velocity TV in the present step is the same as the method of calculating the target velocity TV in S**222**. A method of calculating the target angular velocity Tω in the present step is the same as the method of calculating the target angular velocity Tω in S**232**.
[0269] In S**242**, the main control device **60** causes the cart **402** to turn left by using the target velocity TV and the target angular velocity Tω. When S**242** ends, the process returns to S**210**.
[0270] As mentioned above, the main control device **60** can accurately determine whether the UWB tag **200** is located to the left or right to the center of the cart **402** in the left-right direction by using the fourth distance D4 and the fifth distance D5.
[0271] Some features of the cart **402** disclosed by the present reference example will be listed as below.
(Feature 3-1)
[0272] A cart comprising: [0273] a body unit; [0274] a ground contact unit supported by the body unit and configured to be in contact with a ground; [0275] a prime mover configured to drive the ground contact unit; [0276] a first UWB anchor configured to receive a beacon signal from a UWB tag configured to be carried by a user; [0277] a second UWB anchor arranged to the left of the first UWB anchor and configured to receive the beacon signal; and [0278] a control unit configured to execute a following operation of causing the cart to follow the UWB tag by driving the prime mover, [0279] wherein the control unit is configured to: [0280] calculate a first tag distance which is a distance between the first UWB anchor and the UWB tag by using the beacon signal received by the first UWB anchor, [0281] calculate a second tag distance which is a distance between the second UWB anchor and the UWB tag by using the beacon signal received by the second UWB anchor, and [0282] determine whether the UWB tag is located to the right of the cart, or the UWB tag is located to the left of the cart based on the first tag distance and the second tag distance.
(Feature 3-2)
[0283] The cart according to feature 3-1, wherein the control unit determines a target angular velocity of the cart based on an absolute value of a difference between the first tag distance and the second tag distance.
(Feature 3-3)

[0284] The cart according to feature 3-1 or 3-2, wherein the control unit determines a target velocity of the cart based on an average value of the first tag distance and the second tag distance.
[0285] Effects brought by the above-mentioned features 3-1 to 3-3 will be described.
[0286] In one or more embodiments, the cart **402** comprises: the body unit **4**; the wheels **10** (example for a ground contact unit) supported by the body unit **4** and configured to be in contact with a ground; the traction motor **140** (example for a prime mover) configured to drive the wheels **10**; the right-front UWB anchor **412** (example for a first UWB anchor) configured to receive the beacon signal from the UWB tag **200** configured to be carried by a user; the left-front UWB anchor **410** (example for "second UWB anchor") arranged to the left of the right-front UWB anchor **412** and configured to receive the beacon signal; and the control unit configured to execute a following operation of causing the cart **402** to follow the UWB tag **200** by driving the traction motor **140**. The control unit is configured to: calculate the fourth distance D4 (example for "first tag distance") between the right-front UWB anchor **412** and the UWB tag **200** by using the beacon signal received by the right-front UWB anchor **412**, calculate the fifth distance D5 (example for "second tag distance") between the left-front UWB anchor **410** and the UWB tag **200** by using the beacon signal received by the left-front UWB anchor **410**, and determine whether the UWB tag **200** is located to the right of the cart **402**, or the UWB tag **200** is located to the left of the cart **402** based on the fourth distance D4 and the fifth distance D5.
[0287] According to the above configuration, by using the right-front UWB anchor **412** and the left-front UWB anchor **410** with different positions in the left-right direction, the position of the UWB tag **200** in the left-right direction, i.e., the position of the user in the left-right direction can be suitably determined.
[0288] In one or more embodiments, the control unit determines the target angular velocity Tω of the cart based on the absolute value of the difference between the fourth distance D4 and the fifth distance D5.
[0289] According to the above configuration, the angular velocity of the cart **402** can be adjusted according to the distance between the cart **402** and the user.
[0290] In one or more embodiments, the control unit determines the target velocity TV of the cart **402** based on the average value of the fourth distance D4 and the fifth distance D5.
[0291] According to the above configuration, the velocity of the cart **402** can be adjusted according to the distance between the cart **402** and the user.
[0292] (Modification 3-1) At least one of the right-front UWB anchor **412** and the left-front UWB anchor **410** may comprise one antenna only.
[0293] (Modification 3-2) Positions of the right-front UWB anchor **412** and the left-front UWB anchor **410** in the front-rear direction may differ from each other.
[0294] (Modification 3-3) The right-front UWB anchor **412** and the left-front UWB anchor **410** may be arranged rearward of the load platform **6**.
[0295] (Modification 3-4) In S**222** of FIG. **19**, the main control device **60** may calculate the target velocity TV by using one of the fourth distance D4 and the fifth distance D5. Further, in S**232**, S**240**, the main control device **60** may calculate the target velocity TV and the target angular velocity Tω by using one of the fourth distance D4 and the fifth distance D5.
[0296] (Second Embodiment: Transport System **501**) As illustrated in FIGS. **22** to **25**, a transport system **501** comprises a cart **502** and a beacon **582**.
[0297] (Configuration of Cart **502**) The cart **502** illustrated in FIG. **22** comprises a body **504**, a load platform **506**, a handle **508**, a right-front wheel **510**, a left-front wheel **512**, a right-rear wheel **514**, and a left-rear wheel **516**. The load platform **506**, the handle **508**, the right-front wheel **510**, the left-front wheel **512**, the right-rear wheel **514**, and the left-rear wheel **516** are all supported by the body **504**. The cart **502** transports a load carried on the load platform **506**. The cart **502** comprises a communication module **518** (see FIG. **23**) installed in the body **504**. The cart **502** is configured to operate in a manual mode, a following mode, or a parking mode. In the manual mode, the cart **502**

moves frontward or rearward according to an operation by the user with the user standing behind the body **504** grasping the handle **508** with his/her hands. In the following mode, the cart **502** executes a following operation of moving to follow the beacon **582** (see FIG. **24**) which the user standing in front of the body **504** is carrying. In this case, the cart **502** communicates with the beacon **582** via the communication module **518**. In the parking mode, the cart **502** continues to remain stopped without receiving a command from the handle **508** or receiving a command from the beacon **582**.

[0298] The cart **502** comprises a battery receptacle part **520** arranged in the body **504**. The battery receptacle part **520** is configured to have a battery pack **522** (see FIG. **23**) detachably attached thereto. The battery pack **522** comprises for example secondary battery cell(s) (not illustrated) such as lithium-ion battery cell(s), and is rechargeable by using a charger (not illustrated). The cart **502** operates on power supplied from the battery pack **522** attached to the battery receptacle part **520**.

[0299] As illustrated in FIG. **23**, the cart **502** comprises a right-front wheel motor **524** configured to drive the right-front wheel **510**, a left-front wheel motor **526** configured to drive the left-front wheel **512**, a right-rear wheel motor **528** configured to drive the right-rear wheel **514**, and a left-rear wheel motor **530** configured to drive the left-rear wheel **516**. The right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** are for example brushless motors. The right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** are supported by the body **504**.

[0300] As illustrated in FIG. **22**, the handle **508** is configured to pivot about a pivot axis extending in the up-down direction relative to the body **504**. The user can operate to pivot the handle **508** while the user is grasping the handle **508** with his/her hands. As illustrated in FIG. **23**, the cart **502** comprises a handle angle sensor **532** configured to detect a pivoting angle of the handle **508** as a handle angle, a steering mechanism **534** configured to steer the right-front wheel **510** and the left-front wheel **512** as steering wheels, and a steering motor **536** configured to drive the steering mechanism **534**. The steering motor **536** is for example a brushless motor. The handle angle sensor **532**, the steering mechanism **534**, and the steering motor **536** are supported by the body **504**.

[0301] As illustrated in FIG. **22**, switch boxes **538***a*, **538***b* are disposed on the handle **508**. As illustrated in FIG. **23**, the switch boxes **538***a*, **538***b* include a main power switch **540**, a mode switch **542**, a trigger switch **544**, a moving direction switch **546**, and a velocity switch **548**. The main power switch **540** is configured to switch ON/OFF of main power of the cart **502**. The mode switch **542** is configured to switch the operation mode of the cart **502** between the manual mode, the following mode, and the parking mode. The trigger switch **544** is configured to switch the cart **502** between moving (ON) and not moving (OFF) and/or to adjust a moving velocity of the cart **502** in the manual mode. The moving direction switch **546** is configured to switch a moving direction of the cart **502** between a forward direction and a reverse direction in the manual mode. The velocity switch **548** is configured to switch the moving velocity of the cart **502** between a low speed state and a high speed state in the manual mode. The user can operate the main power switch **540**, the mode switch **542**, the trigger switch **544**, the moving direction switch **546**, and/or the velocity switch **548** while the user is grasping the handle **508** with his/her hands.

[0302] The cart **502** comprises a control power circuit **550** and a control device **552**. The control power circuit **550** permits to supply power from the battery pack **522** to respective components of the cart **502** (e.g., the control device **552**, the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, the left-rear wheel motor **530**, the steering motor **536**) when an ON operation is performed on the main power switch **540**, and prohibits the supply of power from the battery pack **522** to the respective components of the cart **502** when an OFF operation is performed on the main power switch **540**. The control power circuit **550** adjusts the power supplied from the battery pack **522** to a voltage suitable for the respective components of the cart **502** and outputs the same to the respective components of the cart **502**. Also, the control device **552** is composed of a CPU, a ROM, and a RAM, for example. The control device **552** controls operation of the cart **502** by the CPU executing process(es) based on information stored in the ROM and RAM. The control device **552** controls operations of the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, the left-rear wheel motor **530**, and the steering motor **536** via motor drivers **554**, **556**, **558**, **560**, and **562**. Here, the motor drivers **554**, **556**, **558**, and **560** are connected with brake circuits **564**, **566**, **568**, and **570** corresponding to the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530**. The control device **552** can exert a great braking force on the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** by applying a great current on the brake circuits **564**, **566**, **568**, and **570** while the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** are rotating. The control power circuit **550**, the control device **552**, the motor drivers **554**, **556**, **558**, **560**, and **562**, and the brake circuits **564**, **566**, **568**, and **570** are supported by the body **504**.

[0303] (Configuration of Beacon **582**) The beacon **582** illustrated in FIG. **24** is a communication terminal configured to communicate with the cart **502**. The beacon **582** comprises a housing **584**, a main power switch **586**, an initialization switch **588**, a pairing switch **590**, a following start switch **592**, and a following stop switch **594**. The housing **584** has a substantially cuboid shape. A rear surface of the housing **584** comprises a clip part **596**. The beacon **582** can be put on a user's body (e.g., waist belt) via the clip part **596**. Further, a top portion of the housing **584** comprises a ring part **598**. The beacon **582** can be held on a holder (e.g., carabiner) worn by the user via the ring part **598**. The main power switch **586** is arranged at a position on a front surface of the housing **584** and offset to an upper and left side from a center of the housing **584**. The initialization switch **588** is arranged at a position on the front surface of the housing **584** and offset to a lower and right side from the center of the housing **584**. The pairing switch **590** is arranged on a right surface of the housing **584**. The following start switch **592** and the following stop switch **594** are arranged near and at the center of the front surface of the housing **584**. The following start switch **592** is arranged above the following stop switch **594**. A surface of the following start switch **592** comprises projections **600**, and thereby has bumps and dips. For example, the projections **600** are constituted of a plurality of projected bars extending in the left-right direction. Contrary to this, a surface of the following stop switch **594** is smooth. Due to this, the user can distinguish between the following start switch **592** and the following stop switch **594** by confirming presence/absence of the projections **600**. The front-rear direction, the left-right direction, and the up-down direction illustrated in FIG. **24** are different from the front-rear direction, the left-right direction, and the up-down direction which are set with the cart **502** as a reference (see FIG. **22**).

[0304] As illustrated in FIG. **25**, the beacon **582** comprises a microcontroller **602**, a power IC **604**, a charging IC **606**, a battery protection IC **608**, a battery interface **610**, and a USB port **612**. Although details thereof will be described later, the beacon **582** operates on power supplied from a battery **628** or the USB port **612**.

[0305] A first power transmission path **614**, a second power transmission path **616**, and a third power transmission path **618** are arranged between the microcontroller **602**, the battery interface **610**, and the USB port **612**. One end of the first power transmission path **614** is connected to the battery interface **610**. One end of the second power transmission path **616** is connected to the microcontroller **602**. One end of the third power transmission path **618** is connected to the USB port **612**. The other end of the first power transmission path **614**, the other end of the second power transmission path **616**, and the other end of the third power transmission path **618** are connected to each other at a connection point **620**.

[0306] On the first power transmission path **614**, a fuse **622**, a battery protection switch circuit **624**, and a first charging switch circuit **626** of the charging IC **606** are arranged in this order from the battery interface **610** toward the connection point **620**.

[0307] The battery **628** is attached to the battery interface **610**. The battery **628** is for example a

coin-shaped rechargeable secondary battery (e.g., lithium-ion battery).
[0308] The battery protection switch circuit **624** of the present embodiment is a N-channel type MOSFET (Metal-Oxide-Semiconductor Field Effect Transistor), in which a direction from drain to source is a direction from the battery interface **610** to the connection point **620**. Due to this, the battery protection switch circuit **624** permits the current to flow from the battery interface **610** toward the connection point **620** when the battery protection switch circuit **624** is ON, and prohibits the current from flowing from the battery interface **610** toward the connection point **620** when the battery protection switch circuit **624** is OFF. Also, a parasitic diode is formed in the battery protection switch circuit **624**. This parasitic diode permits the current to flow from the source to the drain in the MOSFET. Due to this, in the battery protection switch circuit **624**, no matter whether the battery protection switch circuit **624** is ON/OFF, the current is permitted to flow from the connection point **620** toward the battery interface **610**.
[0309] The first charging switch circuit **626** of the present embodiment is a P-channel type MOSFET, in which the direction from source to drain is a direction from the connection point **620** toward the battery interface **610**. Due to this, the first charging switch circuit **626** permits the current to flow from the connection point **620** toward the battery interface **610** when the first charging switch circuit **626** is ON, and prohibits the current from flowing from the connection point **620** toward the battery interface **610** when the first charging switch circuit **626** is OFF. Also, a parasitic diode is formed in the first charging switch circuit **626**. This parasitic diode permits the current to flow from the drain to the source of the MOSFET. Due to this, in the first charging switch circuit **626**, no matter whether the first charging switch circuit **626** is ON/OFF, the current is permitted to flow from the battery interface **610** toward the connection point **620**.
[0310] The battery protection IC **608** is configured to detect whether an abnormality is occurring in the battery **628**, and switch ON/OFF of the battery protection switch circuit **624** and/or the first charging switch circuit **626** based on its detection result. For example, when a temperature of the battery **628** is an excessively high temperature (e.g., temperature of 60 degrees or higher), the battery protection IC **608** switches the battery protection switch circuit **624** and the first charging switch circuit **626** OFF. Due to this, charging of the battery **628** and power discharge from the battery **628** are prohibited. Alternatively, when the battery **628** is overly discharging electricity, the battery protection IC **608** switches the battery protection switch circuit **624** OFF. Due to this, power discharge from the battery **628** is prohibited. Alternatively, when the battery **628** is overly charging electricity, the battery protection IC **608** switches the first charging switch circuit **626** OFF. Due to this, charging of the battery **628** is prohibited. Although details will be described later, the battery protection IC **608** is also configured to switch the first charging switch circuit **626** between ON/OFF based on a signal output from the microcontroller **602**.
[0311] On the second power transmission path **616**, a regulator **630** of the power IC **604** and a power switch circuit **632** of the power IC **604** are arranged in this order from the microcontroller **602** toward the connection point **620**.
[0312] The microcontroller **602** is composed of, a CPU, a ROM, a RAM for example. The microcontroller **602** controls operation of the beacon **582** by the CPU executing process(es) based on information stored in the ROM and RAM, for example. Also, the microcontroller **602** includes a communication module **634** for communicating with the cart **502**. The communication module **634** is configured to execute pairing with the communication module **518** (see FIG. **23**) of the cart **502** when the pairing switch **590** is operated. By having executed the pairing beforehand, the communication modules **518**, **634** can automatically attempt to establish communication with a pairing counterpart when the power is turned on. Here, the communication modules **518**, **634** each conform to the UWB (Ultra Wide Band) standard. Due to this, communication conforming to the UWB standard is possible between the cart **502** and the beacon **582**.
[0313] The regulator **630** adjusts a supplied power to a predefined control voltage (e.g., 3V) when power is supplied from the battery **628** or the USB port **612**, and outputs the same to the

microcontroller **602**. The power as adjusted to the control voltage is outputted to the microcontroller **602**, by which the microcontroller **602** can operate.

[0314] The power switch circuit **632** of the present embodiment is a P-channel type MOSFET, in which the direction from source to drain is a direction from the connection point **620** toward the microcontroller **602**. Due to this, the power switch circuit **632** permits the current to flow from the connection point **620** toward the microcontroller **602** when the power switch circuit **632** is ON, and prohibits the current from flowing from the connection point **620** toward the microcontroller **602** when the power switch circuit **632** is OFF. Also, a parasitic diode is formed in the power switch circuit **632**. This parasitic diode permits the current to flow from the drain to the source of the MOSFET. Due to this, in the power switch circuit **632**, no matter whether the power switch circuit **632** is ON/OFF, the current is permitted to flow from the microcontroller **602** toward the connection point **620**.

[0315] On the third power transmission path **618**, a fuse **636** and a second charging switch circuit **638** of the charging IC **606** are arranged in this order from the USB port **612** toward the connection point **620**.

[0316] The second charging switch circuit **638** of the present embodiment is a N-channel type MOSFET (Metal-Oxide-Semiconductor Field Effect Transistor), in which a direction from drain to source is a direction from the USB port **612** to the connection point **620**. Due to this, the second charging switch circuit **638** permits the current to flow from the USB port **612** toward the connection point **620** when the second charging switch circuit **638** is ON, and prohibits the current from flowing from the USB port **612** toward the connection point **620** when the second charging switch circuit **638** is OFF. Also, a parasitic diode is formed in the second charging switch circuit **638**. This parasitic diode permits the current to flow from the source to the drain in the MOSFET. Due to this, in the second charging switch circuit **638**, no matter whether the second charging switch circuit **638** is ON/OFF, the current is permitted to flow from the connection point **620** toward the USB port **612**.

[0317] Although not illustrated, the USB port **612** is arranged on an outer surface of the housing **584** (see FIG. **24**). The USB port **612** is configured to receive a USB cable (not illustrated). Here, when the USB cable connected to an external power supply (e.g., commercial power supply) is connected to the USB port **612**, power is supplied from the external power supply via the USB cable and the USB port **612** to the beacon **582**. In this case, the beacon **582** can operate on power supplied from the USB port **612**. In the present embodiment, a state of the beacon **582** being operating on power supplied from the USB port **612** will be termed “external-power-supply operating state”. When the beacon **582** is in the external-power-supply operating state, the power supplied from the external power supply causes a voltage to be applied on each gate of the second charging switch circuit **638** and the power switch circuit **632**, by which each of the second charging switch circuit **638** and the power switch circuit **632** is switched ON. In this case, power is permitted to be supplied from the USB port **612** via the third power transmission path **618** and the second power transmission path **616** to the microcontroller **602**, as a result of which the power from the external power supply is supplied to the microcontroller **602**. Due to this, the microcontroller **602** is activated, and thus the respective components of the beacon **582** can operate (e.g., the battery protection IC **608**).

[0318] When the beacon **582** is in the external-power-supply operating state, the microcontroller **602** switches ON/OFF of the first charging switch circuit **626** according to the remaining level of the battery **628** (i.e., voltage of the battery **628**). For example, when the remaining level of the battery **628** is low, the microcontroller **602** switches the first charging switch circuit **626** ON. Due to this, the first charging switch circuit **626** and the second charging switch circuit **638** are turned ON, by which power is permitted to be supplied from the USB port **612** via the third power transmission path **618** and the first power transmission path **614** to the battery **628**. As a result of this, power from the external power supply is supplied to the battery **628**, by which the battery **628** is charged. Then once the remaining level of the battery **628** has restored to a predefined level, the microcontroller **602** switches the first charging switch circuit **626** OFF. Due to this, power is prohibited from being supplied from the USB port **612** via the third power transmission path **618** and the first power transmission path **614** to the battery **628**, by which charging of the battery **628** ends. Also, when the beacon **582** is in the external-power-supply operating state, the microcontroller **602** switches ON/OFF of the power of the communication module **634** in response to the main power switch **586** being operated. In this case, with the power switch circuit **632** maintained ON, power from the external power supply continues to be supplied to the microcontroller **602**. That is, the “main power” of the beacon **582** herein mentioned is a power supply put into the communication module **634**. Also, when the beacon **582** is in the external-power-supply operating state, the microcontroller **602** outputs an power discharge permitting signal to the battery protection IC **608**. When the power discharge permitting signal is outputted from the microcontroller **602**, the battery protection IC **608** switches the battery protection switch circuit **624** ON. Due to this, the power discharge from the battery **628** is permitted. When the USB cable is removed from the USB port **612** with the battery protection switch circuit **624** ON and thus power supply from the external power supply into the beacon **582** is lost, the beacon **582** starts to operate on power supplied from the battery **628**. In the present embodiment, the state of the beacon **582** at this time will be termed “internal-power-supply operating state”.

[0319] When the beacon **582** is in the internal-power-supply operating state, in response to the main power switch **586** being operated, ON/OFF of the power switch circuit **632** is switched. Due to this, permission/prohibition of the power supply from the battery **628** to the microcontroller **602** is switched, and accordingly operation/prohibition of the microcontroller **602** is switched. Also, when the beacon **582** is in the internal-power-supply operating state, in response to the initialization switch **588** being operated, the microcontroller **602** outputs an power discharge prohibiting signal to the battery protection IC **608**. When the power discharge prohibiting signal is outputted from the microcontroller **602**, the battery protection IC **608** switches the battery protection switch circuit **624** OFF. By the battery protection switch circuit **624** being turned OFF, the power discharge from the battery **628** is prohibited, and accordingly operations of the respective components of the beacon **582** (e.g., the microcontroller **602**) are stopped. In the present embodiment, the state of the beacon **582** at this time will be termed “power supply shutdown state”. In order to switch the battery protection switch circuit **624** ON and thus permit the power discharge from the battery **628** after the beacon **582** has fallen in the power supply shutdown state, the USB cable needs to be connected to the USB port **612**, by which the microcontroller **602** needs to be activated by the power supplied from the external power supply.

[0320] (Usage Example of Initialization Switch **588**) The beacon **582** may be stored in a storage facility as stock for a long period of time (e.g., more than one year) since production before the beacon **582** is delivered to a user. Also, the user may not use the beacon **582** for a long period of time even after obtaining the beacon **582**. During these periods, the beacon **582** may be left without the battery **628** being charged. In this case, because the battery **628** naturally discharges electricity, and the battery **628** may fall into a state of overly discharging electricity. In regards to this phenomenon, the beacon **582** of the present embodiment can suppress the natural power discharge from the battery **628** by operating the initialization switch **588** such that the battery protection switch circuit **624** is turned OFF. Due to this, even if the beacon **582** is left unused for a long period of time without the battery **628** being charged, the battery **628** can be suppressed from falling into the state of overly discharging electricity.

[0321] (Process when Main Power of Beacon **582** is ON: FIG. **26**) When the main power of the beacon **582** (i.e., power of the communication module **634**) is ON, the microcontroller **602** executes processes shown in FIG. **26**.

[0322] In S**302**, the microcontroller **602** switches a beacon following flag OFF. The beacon following flag herein mentioned is information stored in the microcontroller **602** and takes a value of either ON or OFF. After S**302**, the process proceeds to S**304**.

[0323] In S**304**, the microcontroller **602** sends the beacon following flag to the cart **502**. After S**304**, the process proceeds to S**306**.

[0324] In S**306**, the microcontroller **602** assesses whether response information from the cart **502** has been received within a first predetermined time (e.g., 50 milliseconds) since S**306** has started. When the cart **502** receives the beacon following flag sent by the beacon **582**, the cart **502** is configured to send the response information as its response to the beacon **582** (see S**340** in FIG. **27** to be described later). The response information includes mode information indicating the operation mode of the cart **502** and a cart following flag. The cart following flag herein mentioned is information stored in the control device **552** of the cart **502** (see FIG. **23**), and takes either value of ON or OFF. The cart following flag can be regarded as information indicating whether the following operation is in execution in the cart **502** or not. When the first predetermined time has elapsed since S**306** started but the response information is still not received (in case of NO), the process proceeds to S**308**.

[0325] In S**308**, the microcontroller **602** determines that the radio wave was interrupted between the cart **502** and the beacon **582** and thereby communication between the cart **502** and the beacon **582** was not properly executed. After S**308**, the process returns to S**304**.

[0326] When the response information is received within the first predetermined time since S**306** started (in case of YES to S**306**), the process proceeds to S**310**. In S**310**, the microcontroller **602** updates the beacon following flag by using the cart following flag included in the response information received in S**306**. Specifically, when the cart following flag received in S**306** indicates ON, the microcontroller **602** switches the beacon following flag ON, whereas when the cart following flag received in S**306** indicates OFF, the microcontroller **602** switches the beacon following flag OFF. After S**310**, the process proceeds to S**312**.

[0327] In S**312**, the microcontroller **602** assesses whether the following mode is selected in the cart **502** based on the mode information included in the response information received in S**306**. When the following mode is not selected in the cart **502** (in case of NO), the process returns to S**302**. When the following mode is selected in the cart **502** (in case of YES), the process proceeds to S**314**.

[0328] In S**314**, the microcontroller **602** assesses whether the following start switch **592** (see FIG. **24**) is operated or not. When the following start switch **592** is operated (in case of YES), the process proceeds to S**316**.

[0329] In S**316**, the microcontroller **602** switches the beacon following flag ON.

[0330] When in S**314** the following start switch **592** is not operated (in case of NO), or after S**316**, the process proceeds to S**318**. In S**318**, the microcontroller **602** assesses whether the following stop switch **594** (see FIG. **24**) is operated or not. When the following stop switch **594** is operated (in case of YES), the process proceeds to S**320**.

[0331] In S**320**, the microcontroller **602** switches the beacon following flag OFF.

[0332] In S**322**, the microcontroller **602** sends the beacon following flag to the cart **502**. After S**322**, the process proceeds to S**324**.

[0333] In S**324**, the microcontroller **602** assesses whether the microcontroller **602** has received the response information from the cart **502** within the first predetermined time since S**324** started or not. As mentioned above, the response information includes the mode information indicating the operation mode of the cart **502** and the cart following flag. When the first predetermined time has elapsed since S**324** started but still the response information is not received (in case of NO), the process proceeds to S**308**. When the response information is received within the first predetermined time since S**324** started (in case of YES), the process returns to S**312**.

[0334] (Process when Main Power of Cart **502** is ON: FIG. **27**) When the main power of the cart **502** is ON, the control device **552** of the cart **502** (see FIG. **23**) executes processes shown in FIG. **27**.

[0335] In S**332**, the control device **552** switches the cart following flag OFF. As mentioned above, the cart following flag is information stored in the control device **552**, and takes either value of ON or OFF. After S**332**, the process proceeds to S**334**.
[0336] In S**334**, when the following operation control process (see FIG. **28**) is in execution, the control device **552** ends the following operation control process. Although details will be described later, the following operation control process is a process of causing the cart **502** to execute the following operation. When the following operation control process ends and stops being executed, the cart **502** becomes unable to execute the following operation. That is, in S**334**, the following operation by the cart **502** is stopped. After S**334**, the process proceeds to S**336**.
[0337] In S**336**, the control device **552** assesses whether the control device **552** has received the beacon following flag (see S**304**, S**322** in the processes shown in FIG. **26**) sent from the beacon **582** within a second predetermined time (e.g., 5 seconds) since S**336** started or not. When the second predetermined time has elapsed since S**336** started but still the beacon following flag is not received (in case of NO), the process proceeds to S**338**.
[0338] In S**338**, the control device **552** determines that communication between the cart **502** and the beacon **582** is lost. After S**338**, the process returns to S**332**.
[0339] When the beacon following flag is received within the second predetermined time since S**336** started (in case of YES to S**336**), the process proceeds to S**340**. In S**340**, the response information is sent to the beacon **582**. As aforementioned, the response information includes the mode information indicating the operation mode of the cart **502** and the cart following flag. After S**340**, the process proceeds to S**342**.
[0340] In S**342**, whether the following mode is selected in the cart **502** or not is assessed. When the following mode is not selected in the cart **502** (in case of NO), the process returns to S**332**. When the following mode is selected in the cart **502** (in case of YES), the process proceeds to S**344**.
[0341] In S**344**, the control device **552** assesses whether the beacon following flag received in S**336** is ON or not. When the beacon following flag received in S**336** is ON (in case of YES), the process proceeds to S**346**.
[0342] In S**346**, the control device **552** switches the cart following flag ON. After S**346**, the process proceeds to S**348**.
[0343] In S**348**, when the following operation control process (see FIG. **28**) is not in execution, the control device **552** starts the following operation control process. Although details will be described later, by the following operation control process being executed, the cart **502** becomes able to execute the following operation. After S**348**, the process returns to S**336**.
[0344] When the beacon following flag received in S**336** is OFF (in case of NO to S**344**), the process proceeds to S**350**. In S**350**, the control device **552** switches the cart following flag OFF. After S**350**, the process proceeds to S**352**.
[0345] In S**352**, when the following operation control process (see FIG. **28**) is in execution, the control device **552** ends the following operation control process. Although details will be described later, when the following operation control process ends and stops being executed, the cart **502** becomes unable to execute the following operation. That is, in S**352**, the following operation by the cart **502** is stopped. After S**352**, the process returns to S**336**.
[0346] (Following Start Instruction/Following Stop Instruction from Beacon **582** to Cart **502**) According to the processes of FIG. **27**, when the beacon following flag indicating ON is sent from the beacon **582** to the cart **502**, the following operation by the cart **502** is started. Contrary to this, when the beacon following flag indicating OFF is sent from the beacon **582** to the cart **502**, the following operation by the cart **502** is stopped. Due to this, in the present embodiment, the beacon following flag indicating ON is called a following start instruction from the beacon **582** to the cart **502**. Also, the beacon following flag indicating OFF is also called a following stop instruction from the beacon **582** to the cart **502**.
[0347] According to the processes shown in FIG. **26**, immediately after the power of the beacon

582 is ON, irrespective of the operation on the beacon **582**, the beacon following flag indicating OFF (following stop instruction) is sent to the cart **502** (see S**302**, S**304** in FIG. **26**). Accordingly, it is not until the following start switch **592** (see FIG. **24**) is operated after the power of the beacon **582** is turned ON that the beacon following flag indicating ON (following start instruction) is sent to the cart **502**. Due to this, it is possible to suppress the following operation by the cart **502** from being started at a timing unexpected by the user.

[0348] According to the processes shown in FIG. **26**, during a period when communication between the cart **502** and the beacon **582** is established and also the cart **502** is in the following mode (while the processes from S**312** to S**324** are repeatedly executed), operation on the following start switch **592** (see FIG. **24**) and operation on the following stop switch **594** (see FIG. **24**) are received. Due to this, in the present embodiment, the period when communication between the cart **502** and the beacon **582** is established and also the cart **502** is in the following mode is also called “start-operation receiving period”, and also called “stop-operation receiving period”. In the start-operation receiving period (stop-operation receiving period), once the following start switch **592** (the following stop switch **594**) is operated, until the next time the following stop switch **594** (the following start switch **592**) is operated, the beacon **582** continuously sends the following start instruction (following stop instruction) to the cart **502**. Due to this, a situation where the user has operated the following start switch **592** (the following stop switch **594**) but the following operation by the cart **502** is not started (stopped) at all, can be suppressed from occurring.

[0349] (Following Operation Control Process: FIG. **28**) The following operation control process is a process of controlling the following operation of the cart **502**. The following operation control process is executed in the above-mentioned processes of FIG. **27**. Specifically, the following operation control process is started when the cart **502** receives the beacon following flag indicating ON (following start instruction) from the beacon **582** (see S**344** to S**348** in FIG. **27**). Then, the following operation control process ends when the cart **502** receives the beacon following flag indicating OFF (the following stop instruction) from the beacon **582** or when communication between the cart **502** and the beacon **582** is lost (see S**344**, S**350**, S**352**, S**338**, S**332**, and S**334** in FIG. **27**).

[0350] In S**372** shown in FIG. **28**, the control device **552** specifies a distance d to the beacon **582** and an offset angle θo of the beacon **582** by using positioning technolog (ies) based on the UWB communication such as ToA (Time of Arrival) and/or AoA (Angle of Arrival). As illustrated in FIG. **29**, the offset angle θo herein mentioned is an angle of the cart **502** relative to a forward direction FD. In the present embodiment, a direction clockwise in a top view of the cart **502** is defined as a positive direction of the offset angle θo, and a direction counterclockwise in the top view is defined as a negative direction of the offset angle θo. After S**372**, the process proceeds to S**374**.

[0351] In S**374** shown in FIG. **28**, the control device **552** assesses whether the offset angle θo specified in S**372** is within a predetermined first angular range A1 or not. As illustrated in FIG. **29**, the first angular range A1 is an angular range defined as $-\theta 1 \leq \theta o \leq \theta 1$. A first boundary angle θ1 defining a boundary of the first angular range A1 is for example within a range of 5 degrees to 15 degrees, and is 10 degrees in the present embodiment. When the offset angle θo is within the first angular range A1 (in case of YES), the process proceeds to S**376**.

[0352] In S**376** shown in FIG. **28**, the control device **552** specifies a target angular velocity ω to be provided to the cart **502** based on the offset angle θo specified in S**372**. As illustrated in FIG. **29**, the target angular velocity ω herein mentioned is an angular velocity about an axis extending in the up-down direction. In the present embodiment, a Formula (7) for deriving the target angular velocity ω from the offset angle θo is defined as follows:

[00006] $= k * \quad o$ (7)

Here, ωk in the Formula (7) is a predetermined coefficient.

[0353] In S**374** illustrated in FIG. **28**, when the offset angle θo is not within the first angular range

A1 (in case of NO), the process proceeds to S**378**. In S**378**, the control device **552** assesses whether the offset angle θo specified in S**372** is within a predetermined second angular range A2 or not. As illustrated in FIG. **29**, the second angular range A2 is adjacent to the first angular range A1 and is an angular range defined as −θ2≤θo<−θ1, θ1<θo≤θ**2**. A second boundary angle θ**2** is within a range of 15 degrees to 30 degrees for example, and is 30 degrees in the present embodiment. When the offset angle θo is within the second angular range A2 (in case of YES), the process proceeds to S**380**.

[0354] In S**380** illustrated in FIG. **28**, the control device **552** specifies a corrected angle θa based on the offset angle θo specified in S**372**. When the offset angle θo is a positive value, the corrected angle θa is specified such that θa<θo is satisfied, and when the offset angle θo is a negative value, the corrected angle θa is specified such that θo<θa is satisfied. When the offset angle θo is a positive value, the corrected angle θa in the present embodiment is specified as θa=θ1, and when the offset angle θo is a negative value, the corrected angle θa in the present embodiment is specified as θa=−θ1. After S**380**, the process proceeds to S**382**.

[0355] In S**382**, the control device **552** specifies the target angular velocity ω to be provided to the cart **502** based on the corrected angle θa specified in S**380**. In the present embodiment, by the following Formula (8) in which the offset angle θo is replaced with the corrected angle θa based on the Formula (7) used in S**376**, the target angular velocity ω is specified.

[00007] $= k * \ a$ (8)

By specifying the target angular velocity ω from the Formula (8), as compared to the case where the target angular velocity ω is specified from the above-mentioned Formula (7), the target angular velocity ω with a smaller absolute value can be obtained.

[0356] After S**376** or after S**382**, the process proceeds to S**384**. In S**384**, the control device **552** specifies a target moving-straight velocity V to be provided to the cart **502** based on the distance d to the beacon **582** and the offset angle θo that were specified in S**372**. As illustrated in FIG. **29**, the target moving-straight velocity V herein mentioned is a velocity of moving straight in the forward direction FD of the cart **502**. In the present embodiment, a Formula (9) for deriving the target moving-straight velocity V from the distance d to the beacon **582** and the offset angle θo is defined as follows.

[00008] $V = \min(\mathrm{Vk1} * d - \mathrm{Vc1}, \mathrm{Vc2} - \mathrm{Vk2} * \ o)$ (9)

Here, Vk1, Vk2 in the Formula (9) are predetermined coefficients, and Vc1, Vc2 are constant terms.

[0357] In S**378** illustrated in FIG. **28**, when the offset angle θo is not within the second angular range A2 (in case of NO), i.e., the offset angle θo is within a third angular range A3 (see FIG. **29**) defined as θo<−θ2, θ2<θo, the process proceeds to S**386**. In S**386**, the control device **552** specifies the target angular velocity ω to be provided to the cart **502** as ω=0, and specifies the target moving-straight velocity V to be provided to the cart **502** as V=0.

[0358] After S**384** or after S**386**, the process proceeds to S**388**. In S**388**, the control device **552** controls the cart **502** based on the specified target angular velocity ω and the specified target moving-straight velocity V. Specifically, the control device **552** controls the steering motor **536**, the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** (see FIG. **23**) such that the angular velocity and also the moving-straight velocity of the cart **502** coincide the target angular velocity ω and the target moving-straight velocity V. After S**388**, the process returns to S**372**.

[0359] In the following operation control process, when the offset angle θo is within the third angular range A3 (see FIG. **29**), the control device **552** specifies ω=0, V=0 (see S**378** in FIG. **28**). In this case, the control device **552** stops the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** and thus stops the cart **502** from moving (following operation). In other words, when the offset angle θo is within the

third angular range A3, the cart **502** is prohibited to move (the following operation is prohibited).
[0360] FIG. **30** illustrates a relationship between the offset angle θo and the turning curvature K during the following operation of the cart **502** when the distance d from the cart **502** to the beacon **582** is constant in solid lines. When the offset angle θo is within the first angular range A1 (see FIG. **29**), the target angular velocity ω of the cart **502** is specified from the above-described Formula (7) (see S**374**, S**376** in FIG. **28**). As a result of this, the greater the offset angle θo, the greater the turning curvature K. In the present embodiment, the turning curvature K when the target angular velocity ω is based on the Formula (7) will be termed "normal turning curvature K(1)". Also, when the offset angle θo is within the second angular range A2 (see FIG. **29**), the target angular velocity ω of the cart **502** is specified from the above-mentioned Formula (8) (see S**378**, S**380**, S**382** in FIG. **28**). As a result of this, the turning curvature K is constant irrespective of the value of the offset angle θo. In the second angular range A2, the turning curvature K when the target angular velocity ω is based on the Formula (8) is smaller than the normal turning curvature K(1) (see broken lines in FIG. **30**). In the present embodiment, the turning curvature K when the target angular velocity ω is based on the Formula (8) will be termed "reduced turning curvature K(2)".
[0361] A minimum turning radius of the cart **502** in the following mode is within a range of 4339 mm to 15535 mm for example, and is 7012 mm in the present embodiment. The minimum turning radius of the cart **502** in the following mode can be regarded as a turning radius when the turning curvature K is the reduced turning curvature K(2).
[0362] (Processes of Cart **502** in Manual Mode) When the main power of the cart **502** is ON and also the manual mode is selected, the control device **552** illustrated in FIG. **23** controls the cart **502** based on an operation from the user. Specifically, the control device **552** specifies the target angular velocity ω and the target moving-straight velocity V that are to be provided to the cart **502** based on the handle angle detected by the handle angle sensor **532**, a pulling degree of the trigger switch **544**, and a state of the moving velocity of the cart **502** selected in the velocity switch **548**. Then, the control device **552** controls the steering motor **536**, the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** such that the angular velocity and the moving-straight velocity of the cart **502** coincide the specified target angular velocity ω and the specified target moving-straight velocity V. Here, the minimum turning radius of the cart **502** in the manual mode is smaller than the minimum turning radius of the cart **502** in the following mode, and is 953 mm in the present embodiment.
[0363] (Third Embodiment: Transport System **701**) A transport system **701** has a configuration substantially the same as that of the transport system **501** (see FIGS. **22** to **30**) of the second embodiment. The transport system **701** differs from the transport system **501** of the second embodiment only in that the control device **552** of the cart **502** (see FIG. **23**) executes processes illustrated in FIG. **31**, instead of the processes illustrated in FIG. **28**. The processes illustrated in FIG. **31** are the processes illustrated in FIG. **28** with partial modifications. Hereafter, in the processes illustrated in FIG. **31**, only modified points from the processes illustrated in FIG. **28** will be described.
[0364] In the processes illustrated in FIG. **31**, after S**382**, S**400** is executed. In S**400**, the control device **552** specifies the target moving-straight velocity V to be provided to the cart **502** based on the distance d to the beacon **582** specified in S**372** and the corrected angle θa specified in S**380**. In the present embodiment, by using a following Formula (10) in which the offset angle θo is replaced with the corrected angle θa based on the Formula (9) used in S**384**, the target moving-straight velocity V is specified.
[00009] $V = \min(Vk1 * d - Vc1, Vc2 - Vk2 * \quad a) \quad (10)$
After S**400**, the process proceeds to S**388**.
[0365] (Modifications) The beacon **582** may be replaced with another mobile terminal (e.g.,

smartphone, tablet).
[0366] The prime mover configured to drive the right-front wheel **510**, the left-front wheel **512**, the right-rear wheel **514**, and the left-rear wheel **516** of the cart **502** may be replaced with a prime mover other than the electric motor (e.g., engine comprising an internal combustion mechanism).
[0367] The operation of the cart **502** may not comprise the manual mode (or the parking mode). That is, the operation mode of the cart **502** may be switched between the following mode and the parking mode (or the manual mode). Alternatively, the operation mode of the cart **502** may only comprise the following mode. In this case, the cart **502** may not comprise the mode switch **542**.
[0368] In S**372** of FIG. **28**, the control device **552** may use a positioning technology other than the positioning technologies based on the UWB communication to specify the distance d to the beacon **582** and the offset angle θo of the beacon **582**. For example, a positioning technology based on Bluetooth (registered trademark) communication and a positioning technology based on Wi-Fi (registered trademark) communication may be used.
[0369] In the processes of FIG. **28**, the control device **552** may specify a target steering angle (swiveling angle of steering wheels relative to the forward direction FD) to be provided to the steering wheels of the cart **502** (the right-front wheel **510** and the left-front wheel **512**) instead of specifying the target angular velocity ω to be provided to the cart **502**. In this case, in S**388** of FIG. **28**, the control device **552** may control the steering motor **536**, the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** after adjusting the steering angle of the steering wheels to the target steering angle such that the moving-straight velocity of the cart **502** coincides the target moving-straight velocity V.
[0370] The cart **502** may not comprise the steering mechanism **534**. Even in this case, the cart **502** can perform a turning operation by providing a difference between a rotation speed of the right-front wheel **510** and the right-rear wheel **514** and a rotation speed of the left-front wheel **512** and the left-rear wheel **516**. In S**388** in FIG. **28** for example, the control device **552** of the cart **502** may control the rotation speeds of the right-front wheel **510**, the left-front wheel **512**, the right-rear wheel **514**, and the left-rear wheel **516** such that the angular velocity and the moving-straight velocity of the cart **502** coincide the target angular velocity ω and the target moving-straight velocity V.
[0371] In the processes of FIG. **28**, after NO is determined in S**374**, the process may skip S**378**, and proceed to S**380**. That is, even when the offset angle θo is within the third angular range A3, the control device **552** may not stop the cart **502** from moving (following operation) but rather may continue to allow the cart **502** to move.
[0372] The beacon **582** may not comprise the following stop switch **594** (see FIG. **24**). Even in this case, the user can indirectly stop the following operation by the cart **502** by operating the main power switch **586** (see FIG. **24**) and turning the main power of the beacon **582** OFF. This is because communication between the cart **502** and the beacon **582** is lost due to the main power of the beacon **582** being turned OFF, NO is determined to S**336** in FIG. **27**, and the subsequent processes of S**338**, S**332**, and S**334** cause the cart **502** to stop the following operation.
[0373] A starting instruction send finish condition for finishing sending of the following start instruction from the beacon **582** to the cart **502** may include a condition other than the following stop switch **594** (see FIG. **24**) being operated. For example, the starting instruction send finish condition may include a condition that a predetermined time elapses since the following start switch **592** (see FIG. **24**) is operated. Alternatively, the starting instruction send finish condition may include a condition that the cart **502** has finished the following operation control process (see FIG. **28**).
[0374] A stopping instruction send finish condition for finishing sending the following stop instruction from the beacon **582** to the cart **502** may include a condition other than the following start switch **592** (see FIG. **24**) being operated. For example, the stopping instruction send finish condition may include a condition that a predetermined time elapses after the following stop switch

594 (see FIG. **24**) is operated. Alternatively, the stopping instruction send finish condition may include a condition that the cart **502** has started the following operation control process (see FIG. **28**).

[0375] The start-operation receiving period (the stop-operation receiving period) may include a period during which the cart **502** is in the manual mode (or the parking mode). Due to this, even in the period during which the cart **502** is in the manual mode (or the parking mode), operation on the following start switch **592** (see FIG. **24**) and operation on the following stop switch **594** (see FIG. **24**) may be received.

[0376] The start-operation receiving period (the stop-operation receiving period) may include a period during which communication between the cart **502** and the beacon **582** is not established. Due to this, even in the period during which communication between the cart **502** and the beacon **582** is not established, operation on the following start switch **592** (see FIG. **24**) and operation on the following stop switch **594** (see FIG. **24**) may be received.

[0377] In the processes of FIG. **26**, the process of S**302** may be omitted. That is, immediately after the power of the beacon **582** is turned on, or the following mode is not selected in the cart **502** (in case of NO to S**312**), the beacon following flag may not be switched OFF.

[0378] The battery **628** (see FIG. **25**) may be a primary battery which is not rechargeable (e.g., may be a manganese lithium battery). In this case, the battery **628** may be attachable and detachable from the beacon **582**.

[0379] The battery protection switch circuit **624**, the power switch circuit **632**, the first charging switch circuit **626**, and the second charging switch circuit **638** (see FIG. **25**) may be respectively mechanical switches.

[0380] The beacon **582** may further comprise an operation unit (e.g., switch) for switching the battery protection switch circuit **624** (see FIG. **25**) ON. In this case, in order to switch the battery protection switch circuit **624** from OFF to ON, the above-mentioned operation unit may be required to be operated after setting the beacon **582** in the external-power-supply operating state.

[0381] (Features in Second and Third Embodiments) In one or more embodiments, the cart **502** may be configured to execute the following operation of autonomously following the beacon **582** (example for a following target). The cart **502** comprises: the body **504**; the right-front wheel **510**, the left-front wheel **512**, the right-rear wheel **514**, and the left-rear wheel **516** (example for a wheel) supported by the body **504** and configured to be in contact with a ground; the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** (example for a prime mover) configured to drive the right-front wheel **510**, the left-front wheel **512**, the right-rear wheel **514**, and the left-rear wheel **516**; the communication module **518** and the control device **552** (example for an offset angle detector) configured to detect the offset angle θo of the beacon **582** relative to the forward direction FD of the cart **502**; and the control device **552**. The control device **552** is configured to execute the following operation control process of controlling the following operation of the cart **502**. In the following operation control process, when the offset angle θo is within the first angular range A1 including 0 degrees, the control device **552** adjusts the turning curvature K(example for a turning degree) of the cart **502** during the following operation to the normal turning curvature K(1) corresponding to the offset angle θo, and when the offset angle θo is within the second angular range A2 adjacent to the first angular range A1, the control device **552** adjusts the turning curvature K of the cart **502** during the following operation to the reduced turning curvature K(2) which is reduced from the normal turning curvature K(1) corresponding to the offset angle θo.

[0382] When an absolute value of the offset angle θo is greater, a turning angle required to direct the forward direction FD of the cart **502** toward the beacon **582** is greater, and due to this, increasing the turning curvature K of the cart **502** may be considered. However, if the turning curvature K of the cart **502** is excessively large, an inner wheel difference of the cart **502** may be excessively enlarged. As a result, the cart **502** may collide with an obstacle, as a result of which the

following operation of the cart **502** may not smoothly proceed. According to the above configuration, when the offset angle θo is within the second angular range A2 (that is, the absolute value of the offset angle θo is relatively large), the turning curvature K of the cart **502** is reduced than in normal times. Due to this, the turning curvature K of the cart **502** can be suppressed from becoming excessively large, by which the inner wheel difference of the cart **502** can be suppressed from being excessively enlarged. Due to this, the cart **502** can be suppressed from contacting an obstacle, by which the following operation of the cart **502** can be smoothly proceeded.
[0383] In one or more embodiments, the turning degree includes the turning curvature K of the cart **502**. In the following operation control process, when the offset angle θo is within the first angular range A1, the control device **552** adjusts the turning curvature K of the cart **502** during the following operation to the normal turning curvature K(1) corresponding to the offset angle θo, and when the offset angle θo is within the second angular range A2, the control device **552** adjusts the turning curvature K of the cart **502** during the following operation to the reduced turning curvature K(2) which is reduced from the normal turning curvature K(1) corresponding to the offset angle θo.
[0384] According to the above configuration, when the offset angle θo is within the second angular range A2 (i.e., the absolute value of the offset angle θo is relatively large), the turning curvature K of the cart **502** can be reduced than in normal times. Due to this, the turning curvature K of the cart **502** can be suppressed from being excessively increased, by which the inner wheel difference of the cart **502** can be suppressed from being excessively enlarged. Due to this, since the cart **502** can be suppressed from contacting an obstacle, the following operation of the cart **502** can be smoothly proceeded.
[0385] In one or more embodiments, the cart **502** is configured to switch between the following mode in which execution of the following operation is permitted and the manual mode in which the execution of the following operation is prohibited and the cart **502** moves based on an operation from a user. The minimum turning radius of the cart **502** in the following mode is greater than the minimum turning radius of the cart **502** in the manual mode.
[0386] When the cart **502** has a smaller turning radius, the inner wheel difference of the cart **502** is enlarged. Due to this, in the following mode in which the operation by the user does not intervene, the cart **502** may contact an obstacle if the turning radius of the cart **502** becomes small. In the meantime, in the manual mode in which the operation by the user intervenes, even when the turning radius of the cart **502** becomes small to a certain degree, the possibility of the cart **502** contacting an obstacle is probably low. Rather, if the turning radius of the cart **502** cannot be made small in the manual mode, maneuverability of the cart **502** may be decreased. According to the above configuration, the minimum turning radius of the cart **502** in the following mode is greater than the minimum turning radius of the cart **502** in the manual mode. As a result, the turning radius of the cart **502** can be suppressed from being made small in the following mode, and the turning radius can be permitted to be small in the manual mode. Due to this, the cart **502** can be suppressed from contacting an obstacle without imparting the maneuverability of the cart **502** in the manual mode.
[0387] In one or more embodiments, in the following operation control process, when the offset angle θo is within the third angular range A3 excluding the first angular range A1 and the second angular range A2, the control device **552** stops the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** to stop the following operation by the cart **502**.
[0388] When an absolute value of the offset angle θo is greater and thus a turning angle required to direct the forward direction FD of the cart **502** to the beacon **582** is greater, the cart **502** would need to be turned greatly. However, if the cart **502** is turned greatly, the cart **502** may contact an obstacle, by which the following operation of the cart **502** may not proceed smoothly. According to the above configuration, when the offset angle θo is within the third angular range A3 (i.e., the absolute value of the offset angle θo is relatively large), the following operation by the cart **502** is stopped.

Due to this, the cart **502** can be suppressed from turning greatly. As a result, the cart **502** can be suppressed from contacting the obstacle, the following operation by the cart **502** can be smoothly proceeded.

[0389] In one or more embodiments, the cart **502** is configured to execute a following operation of autonomously following the beacon **582** (example for a following target). The cart **502** comprises: the body **504**; the right-front wheel **510**, the left-front wheel **512**, the right-rear wheel **514**, and the left-rear wheel **516** (example for a wheel) supported by the body **504** and configured to be in contact with a ground; the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** (example for a prime mover) configured to drive the right-front wheel **510**, the left-front wheel **512**, the right-rear wheel **514**, and the left-rear wheel **516**; the communication module **518** and the control device **552** (example for an offset angle detector) configured to detect the offset angle θo of the beacon **582** relative to the forward direction FD of the cart **502**; and the control device **552**. The control device **552** is configured to operate the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** to cause the cart **502** to execute the following operation when the offset angle θo is within the first angular range A1 and the second angular range A2 (example for an operation angular range) including 0 degrees, and the control device **552** is configured to stop the right-front wheel motor **524**, the left-front wheel motor **526**, the right-rear wheel motor **528**, and the left-rear wheel motor **530** to stop the following operation by the cart **502** when the offset angle θo is within the third angular range A3 (example for a stop angular range) adjacent to the first angular range A1 and the second angular range A2.

[0390] When an absolute value of the offset angle θo is greater and thus a turning angle required to direct the forward direction FD of the cart **502** toward the beacon **582** is greater, the cart **502** would need to be turned greatly. However, if the cart **502** is turned greatly, the cart **502** may contact an obstacle, by which the following operation of the cart **502** may not proceed smoothly. According to the above configuration, when the offset angle θo is within the first angular range A1 and the second angular range A2 (i.e., the absolute value of the offset angle θo is relatively small), the following operation by the cart **502** is executed. When the offset angle θo is within the third angular range A3 (i.e., the absolute value of the offset angle θo is relatively large), the following operation by the cart **502** is stopped. Due to this, the cart **502** can be suppressed from turning greatly. As a result, the cart **502** can be suppressed from contacting the obstacle, the following operation by the cart **502** can be smoothly proceeded.

[0391] In one or more embodiments, the transport system **501**, **701** comprises: the cart **502**; and the beacon **582** (example for a communication terminal) configured to communicate with the cart **502**. The cart **502** is configured to execute a following operation of following the beacon **582** when the beacon **582** is moving. The beacon **582** comprises: the main power switch **586** (example for a first operation part); and the following start switch **592** (example for a second operation part) located at a position different from the main power switch **586**. When the main power switch **586** is operated, the beacon **582** switches ON/OFF of a power of the beacon **582**. When the following start switch **592** is operated during the predetermined start-operation receiving period (period during which the processes from S**312** to S**324** in FIG. **26** are repeatedly executed) with the power on, the beacon **582** sends the following start instruction which instructs to start the following operation (beacon following flag indicating ON) to the cart **502**.

[0392] According to the above configuration, the beacon **582** includes the following start switch **592** for causing the cart **502** to start the following operation at the position different from the main power switch **586**. Even if the main power switch **586** is operated without user's intention, unless the following start switch **592** is operated thereafter, the following operation by the cart **502** is not started. Due to this, the following operation by the cart **502** can be suppressed from being started without user's intension.

[0393] In one or more embodiments, the beacon **582** may further comprise the following stop switch **594** (example for a third operation part) disposed at a position different from those of the main power switch **586** and the following start switch **592**. When the following stop switch **594** is operated during the predetermined stop operation receiving period (period during which the processes from S**312** to S**324** in FIG. **26** are repeatedly executed) with the power on, the beacon **582** sends the following stop instruction which instructs to stop the following operation (the beacon following flag indicating OFF) to the cart **502**.

[0394] It is also conceivable to provide the beacon **582** with a switching operation part configured to receive switching between starting and stopping of the following operation. In this configuration however, when the user operates the switching operation part, whether the beacon **582** instructs the cart **502** to start or stop the following operation may be unclear for the user. According to the above configuration, the operation part (following start switch **592**) for receiving the start of the following operation and the operation part (following stop switch **594**) for receiving the stop of the following operation are disposed separate from each other. Due to this, when the user operates the following start switch **592**/the following stop switch **594**, whether the beacon **582** instructs the cart **502** to start/stop the following operation is clarified for the user.

[0395] In one or more embodiments, when the following stop switch **594** is operated during the stop operation receiving period with the power on, the beacon **582** continuously sends the following stop instruction to the cart **502** until the predetermined send finish condition is satisfied.

[0396] For example, if communication connection is not good between the cart **502** and the beacon **582**, the following stop instruction sent from the beacon **582** may not be received at the cart **502**. If the beacon **582** stops sending the following stop instruction despite the cart **502** has not received the following stop instruction, the following operation by the cart **502** would not be stopped, and thus would undesirably continue thereafter. In this configuration, a situation in which the user has operated the following stop switch **594** but the following operation by the cart **502** is not stopped at all may happen. According to the above configuration, after the following stop switch **594** is operated, the beacon **582** continuously sends the following stop instruction until the send finish condition is satisfied. Due to this, the situation in which the user has operated the following stop switch **594** but the following operation by the cart **502** is not stopped at all can be suppressed.

[0397] In one or more embodiments, the send finish condition includes the first send finish condition that the following start switch **592** is operated.

[0398] According to the above configuration, the beacon **582** continuously sends the following stop instruction from when the following stop switch **594** is operated until the following start switch **592** is operated next. Due to this, the situation in which the user has operated the following stop switch **594** but the following operation by the cart **502** is not stopped at all can be suppressed.

[0399] In one or more embodiments, the cart **502** is configured to switch between the following mode in which execution of the following operation is permitted and the manual mode in which the execution of the following operation is prohibited and the cart **502** moves based on an operation from a user. The start-operation receiving period is at least a period during which the cart **502** is in the following mode.

[0400] If a period during which the cart **502** is in the manual mode is the start-operation receiving period, when the following start switch **592** is operated during the manual mode, the following operation by the cart **502** may undesirably start at the timing when the cart **502** is switched to the following mode thereafter. Due to this, the following operation by the cart **502** may undesirably start a while late after the user has operated the following start switch **592**. This means that the following operation by the cart **502** starts at an unexpected timing for the user. According to the above configuration, the start-operation receiving period is at least a period during which the cart **502** is in the following mode. Due to this, even if the following start switch **592** is operated during the manual mode, since that operation is not accepted, the following operation by the cart **502** is not to start thereafter. Due to this, the following operation by the cart **502** can be suppressed from being started at an unexpected timing for the user.

[0401] In one or more embodiments, the start-operation receiving period is at least a period during which communication between the cart **502** and the beacon **582** is established.
[0402] If a period during which the communication between the cart **502** and the beacon **582** is lost is the start-operation receiving period, when the following start switch **592** is operated during such period, the following operation by the cart **502** may undesirably start at a timing when the communication is established thereafter. Due to this, the following operation by the cart **502** may undesirably start a while late after the user has operated the following start switch **592**. This means that the following operation by the cart **502** starts at an unexpected timing for the user. According to the above configuration, the start-operation receiving period is at least a period during which the communication between the cart **502** and the beacon **582** is established. Due to this, even if the following start switch **592** is operated during the period when the communication between the cart **502** and the beacon **582** is lost, since that operation is not accepted, the following operation by the cart **502** is not to start thereafter. Due to this, the following operation by the cart **502** can be suppressed from being started at an unexpected timing for the user.
[0403] In one or more embodiments, the beacon **582** sends the following stop instruction which instructs the cart **502** to stop the following operation to the cart **502** immediately after the power is turned on.
[0404] According to the above configuration, the following operation by the cart **502** is stopped immediately after the power of the beacon **582** is turned on. Due to this, for the first time the following start switch **592** is operated after the power of the beacon **582** is turned on, the following operation by the cart **502** is started. Due to this, the following operation by the cart **502** can be suppressed from being started without a user's intention.
[0405] In one or more embodiments, the beacon **582** (example for electric apparatus) comprises: the microcontroller **602**; the battery interface **610** configured to be electrically connected to the battery **628**; the power switch circuit **632** (example for a first switch circuit) configured to permit power supply to the microcontroller **602** when the power switch circuit **632** is ON and configured to prohibit power supply to the microcontroller **602** when the power switch circuit **632** is OFF; and the battery protection switch circuit **624** (example for a second switch circuit) configured to permit power discharge from the battery **628** when the battery protection switch circuit **624** is ON and prohibit power discharge from the battery **628** when the battery protection switch circuit **624** is OFF.
[0406] When the beacon **582** is left unused for a long period of time with the battery **628** attached to the battery interface **610**, a remaining level of the battery **628** may decrease due to natural power discharge. According to the above configuration, the battery protection switch circuit **624** configured to switch between permission/prohibition on the power discharge from the battery **628** is provided separately from the power switch circuit **632** configured to switch permission/prohibition on the power supply to the microcontroller **602** (i.e., ON/OFF of power). In this configuration, by having the battery protection switch circuit **624** OFF, the natural power discharge from the battery **628** can be suppressed. Due to this, by having the battery protection switch circuit **624** OFF when the beacon **582** is left unused for a long period of time, the decrease in the remaining level of the battery **628** can be suppressed. In one or more embodiments, the battery protection switch circuit **624** and the power switch circuit **632** are arranged in series on the power supply path from the battery interface **610** to the microcontroller **602** (i.e., the first power transmission path **614**, the connection point **620**, and the second power transmission path **616**).
[0407] According to the above configuration, by having the battery protection switch circuit **624** OFF, irrespective of the ON/OFF state of the power switch circuit **632**, the power supply to the microcontroller **602** can be prohibited.
[0408] In one or more embodiments, the power switch circuit **632** is a MOSFET (example for electrical switch). The battery protection switch circuit **624** is a MOSFET (example for electrical switch).

[0409] According to the above configuration, as compared to when the power switch circuit **632** (battery protection switch circuit **624**) is for example a mechanical switch, the power switch circuit **632** (battery protection switch circuit **624**) can be made smaller.
[0410] In one or more embodiments, the beacon **582** further comprises the USB port **612** configured to receive a USB cable. The battery protection switch circuit **624** is switched from OFF to ON when the beacon **582** is supplied with power from the USB cable via the USB port **612**.
[0411] According to the above configuration, with a relatively simple operation of connecting the USB cable to the USB port **612**, the battery protection switch circuit **624** can be switched from OFF to ON.
[0412] In one or more embodiments, the battery **628** is a rechargeable secondary battery.
[0413] When the battery **628** is a secondary battery, the battery **628** may be sometimes configured difficult to be removed from the beacon **582**. Due to this, the battery **628** is stored in the state attached to the beacon **582**, and natural power discharge from the battery **628** may occur while the beacon **582** is stored. In order to address this situation, according to the above configuration, by having the battery protection switch circuit **624** OFF while the beacon **582** is stored, the natural power discharge from the battery **628** can be suppressed.

Claims

1. A cart comprising: a body unit; a ground contact unit supported by the body unit and configured to be in contact with a ground; a prime mover configured to drive the ground contact unit; a first UWB anchor including a first antenna and a second antenna different from the first antenna and configured to receive a beacon signal from a UWB tag configured to be carried by a user; a second UWB anchor a position of which is different from that of the first UWB anchor in a front-rear direction and configured to receive the beacon signal from the UWB tag; and a control device configured to execute a following operation of causing the cart to follow the UWB tag by driving the prime mover, wherein the control device is configured to calculate a first distance which is a distance between the first UWB anchor and the UWB tag and a first tag angle which is an angle of the UWB tag relative to the first UWB anchor by using the beacon signal received by the first antenna and the beacon signal received by the second antenna.
2. The cart according to claim 1, wherein the control device is configured to calculate a second distance which is a distance between the second UWB anchor and the UWB tag by using the beacon signal received by the second UWB anchor, and the control device is configured to determine whether the UWB tag is located in a first area or a second area different from the first area by using the first distance and the second distance.
3. The cart according to claim 2, wherein the first UWB anchor is located frontward of the second UWB anchor, the first antenna and the second antenna are aligned in a left-right direction, a midpoint between the first antenna and the second antenna in the left-right direction matches a center of the second UWB anchor in the left-right direction, when a second tag angle which is an angle between a first virtual line connecting the first UWB anchor and the UWB tag and a second virtual line connecting the first UWB anchor and the second UWB anchor is an obtuse angle, the control device determines that the UWB tag is located in the first area, and when the second tag angle is 90 degrees or an acute angle, the control device determines that the UWB tag is located in the second area.
4. The cart according to claim 2, wherein the first area is a following operation permitted area in which the following operation is permitted, and the second area is a following operation prohibited area in which the following operation is prohibited.
5. The cart according to claim 4, wherein when the control device determines that the UWB tag is located in the following operation permitted area, the control device calculates a target velocity and a target angular velocity by using the first distance and the first tag angle to control the prime

mover.

6. The cart according to claim 5, wherein when the control device determines that the UWB tag has moved from the following operation permitted area to the following operation prohibited area, the control device stops the prime mover, and when the control device determines that the UWB tag has moved from the following operation prohibited area to the following operation permitted area, the control device drives the prime mover.

7. The cart according to claim 1, wherein the first UWB anchor has a higher receiving sensitivity for signals arriving from a front side than for signals arriving from a rear side.

8. The cart according to claim 1, further comprising a load platform supported by the body unit, wherein one of the first UWB anchor and the second UWB anchor is a front UWB anchor located frontward of the load platform.

9. The cart according to claim 8, wherein the other of the first UWB anchor and the second UWB anchor is a rear UWB anchor located rearward of the load platform, and the rear UWB anchor is located above the front UWB anchor and the load platform.

10. The cart according to claim 9, further comprising a handle including a grip configured to be grasped by the user, wherein the handle is located above the load platform, and the rear UWB anchor is disposed in or on the handle.

11. The cart according to claim 3, wherein the first area is a following operation permitted area in which the following operation is permitted, the second area is a following operation prohibited area in which the following operation is prohibited, when the control device determines that the UWB tag is located in the following operation permitted area, the control device calculates a target velocity and a target angular velocity by using the first distance and the first tag angle to control the prime mover, when the control device determines that the UWB tag has moved from the following operation permitted area to the following operation prohibited area, the control device stops the prime mover, when the control device determines that the UWB tag has moved from the following operation prohibited area to the following operation permitted area, the control device drives the prime mover, the first UWB anchor has a higher receiving sensitivity for signals arriving from a front side than for signals arriving from a rear side, the cart further comprises a load platform supported by the body unit, one of the first UWB anchor and the second UWB anchor is a front UWB anchor located frontward of the load platform, the other of the first UWB anchor and the second UWB anchor is a rear UWB anchor located rearward of the load platform, the rear UWB anchor is located above the front UWB anchor and the load platform, the cart further comprises a handle including a grip configured to be grasped by the user, the handle is located above the load platform, and the rear UWB anchor is disposed in or on the handle.